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Title: Enhanced Inverse Synthetic Aperture Radar Imagery Using 2-D Spectral Estimation

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This research focused on the use of classical and modern two-dimensional spectral estimation techniques for enhancing inverse synthetic aperture radar (ISAR) imagery. 2-D Classical spectral estimation methods and 2-D modern parametric and non-parametric spectral estimation methods are applied to simulated and real data sets in order to achieve both range and cross-range resolution enhancement. After a basic introduction of radar fundamentals the concepts of image formation for ISAR imagery is developed from a 2-D matched filter method into 2-D Fourier based methods. The concepts of Zero-Doppler clutter subtraction and polar-to-cartesian interpolation are introduced and applied. 2-D Classical and modern spectral estimation methods are presented and integrated into image formation algorithms. A quantitative and qualitative assessment of the imagery product from each is made based on resolution, variance, and dynamic range. A Matlab toolbox for ISAR image formation of both simulated and measured ISAR data is developed. Alternative image formation techniques such as Hybrid algorithms and Time-Frequency Analysis are introduced.

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# Enhanced Inverse Synthetic Aperture Radar Imagery Using 2-D Spectral Estimation

by

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David R. Ohm, Author

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# ENHANCED INVERSE SYNTHETIC APERTURE RADAR IMAGERY USING 2-D SPECTRAL ESIMATION

## CHAPTER ONE

### 1.1 INTRODUCTION

This research focused on the use of classical and modern two-dimensional spectral estimation techniques for enhancing inverse synthetic aperture radar (ISAR) imagery. We investigate the use of classical spectral estimation methods and modern parametric and non-parametric spectral estimation methods to achieve both range and cross-range resolution enhancement.

This chapter begins with a basic review of radar operation and systems. The concept of *radar imaging* and *resolution* will be presented in section 1.2 along with an introduction to synthetic aperture imaging. A short description of this work including the purpose for this research is given in section 1.3. Finally, this chapter ends with a summary of research contributions.

It is assumed that the reader has a reasonable background and understanding in digital signal processing and radar theory. It is also assumed that the reader is at least familiar with the synthetic aperture concept, though this is reviewed in the first two chapters. For additional reading on any of these subjects, there is a wide assortment of books available. For an excellent treatment of digital signal processing, see [1]. Radar theory has been the topic of many good books, a good example of which is [2]. A selection of books that cover inverse synthetic aperture radar processing can be found in the reference section of this thesis [3-5]. Also, several papers appear in the published literature covering the fundamentals of ISAR imaging [6-8].

## 1.2 RADAR BACKGROUND

A typical radar system consists of an antenna, transmitting subsystem and a receive subsystem, and uses a modulated waveform and directional antenna to transmit electromagnetic radiation into a specific region of space to image or search for targets. These targets, or objects, reflect or backscatter part of that radiation back to the radar system. The reflected radiation makes up the return signal that is stored for off-line processing, or processed in real time [9], in order to extract information about the target range, velocity, reflectivity, and/or other identifying characteristics (see Figure 1.1).

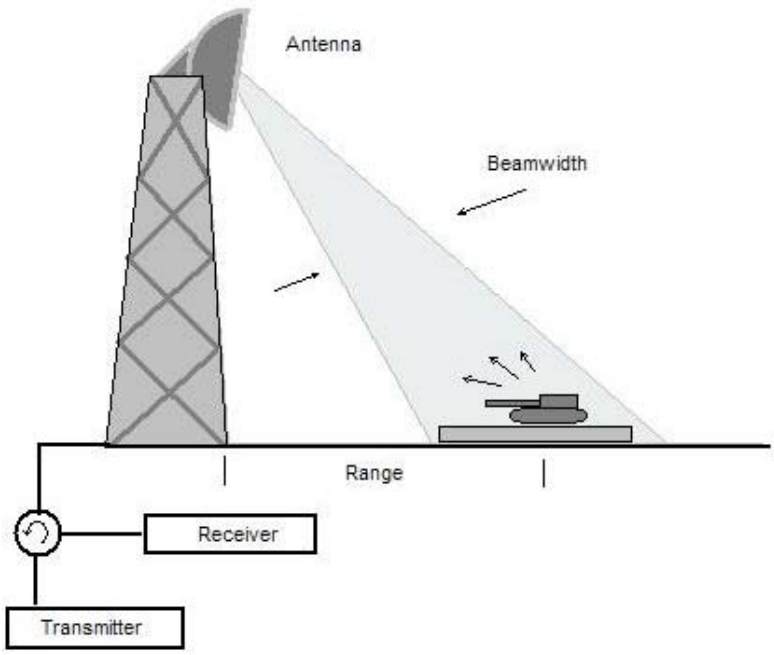



Figure 1.1 Basic radar system showing backscatter from a target

There are several major types of radar systems including: airborne, spaceborne, ship based, and ground based systems. These major radar systems have several applications including: weather monitoring/prediction, searching, tracking, early warning, terrain

avoidance, over-the-horizon, and more. There are also phased array radar systems that are a composite antenna formed from two or more basic radar antennas. Phased array radar systems have an advantage in some applications in that they can produce narrow directive beams that can be steered electronically or mechanically.

Radar systems can be classified by the waveforms they employ. The two main classifications based on waveform are continuous wave (CW) radar and modulated wave radar. Continuous wave radar systems are usually used to measure target radial velocity by Doppler shift techniques. They are also useful for angular position measurements and are often used in missile guidance applications [10]. Modulated waveform radar systems are needed in order to measure target range. Often pulse modulation is used and can be understood as a train of pulses transmitted by the radar with a low, medium or high Pulse Repetition Frequency (PRF). Low PRF radar systems are mainly used to measure target range when target velocity is not of interest. High PRF radar systems are mainly used to measure both target range and velocity.

Radar systems can be further characterized by operating frequency. Table 1.1 shows the many radar classifications based on the operating frequency bands. High frequency (HF) radars can utilize the ionosphere by reflecting electromagnetic waves off of it in order to detect targets beyond the horizon. This is called *over-the-horizon radar*. Very high frequency radar (VHF) and ultra high frequency radar (UHF) bands are used in early warning radar systems and foliage penetrating (FOPEN) synthetic aperture radar systems. It is worth noting that very large antenna apertures are needed for these systems due to the large operating wavelength and the sensitivity required for long-range measurements. Radars in the L-band are primarily used for ground-based systems such as air traffic control search

operations, and satellite based remote sensing systems such as RADARSAT and SRTM [11,12].

TABLE 1.1. Typical radar frequency bands

| Designation | Frequency (GHz) |
|---|---|
| HF | 0.003 – 0.03 |
| VHF | 0.03 – 0.3 |
| UHF | 0.3 – 1.0 |
| L-band | 1.0 – 2.0 |
| S – band | 2.0 – 4.0 |
| C – band | 4.0 – 8.0 |
| X – band | 8.0 – 12.5 |
| Ku – band | 12.5 – 18.0 |
| K – band | 18.0 – 26.5 |
| Ka – band | 26.5 – 40.0 |
| MMW | Normally > 34.0 |

Most medium range radars for ships and ground-based units operate in the S-band, while most weather detection radars operate in the C-band. For applications where the size of the antenna is limited and atmospheric attenuation is a concern, X-band is often the radar operating frequency employed. Higher frequency bands such as Ku, K, and Ka suffer from severe atmospheric attenuation. This attenuation results from absorption by atmospheric molecules or scattering by aerosols in the atmosphere between the microwave sensor on board a

spacecraft or aircraft and the target to be measured. Radars using these bands are often limited to short range applications such as very high-resolution reconnaissance from aircraft or UAVs [9]. Table 1.2 lists radar operating frequency ranges along with common applications within those ranges.

TABLE 1.2 Frequency bands for radar applications. [Knott et al, 1993]

| Frequency Range | General Usage |
|---|---|
| 50-1000 MHz | Very Long-Range Surveillance |
| 1-2 GHz | Long-Range Surveillance,<br>Enroute Traffic Control |
| 2-4 GHz | Moderate Range Surveillance,<br>Terminal Traffic Control,<br>Long-Range Weather |
| 4-8 GHz | Long-Range Tracking,<br>Airborne Weather Detection |
| 8-12 GHz | Short Range Tracking,<br>Missile Guidance,<br>Mapping,<br>Marine Radar,<br>Airborne Intercept |
| 12-18 GHz | High-Resolution Mapping,<br>Satellite Altimetry |
| 18-27 GHz | Little Used (Water Vapor Absorption) |
| 27-40 GHz | Very High Resolution Mapping,<br>Airport Surveillance |
| 40-100+ GHz | Experimental |

## 1.3 RADAR IMAGING

In order to characterize the scattering properties of a target it is possible to create a spatial distribution of reflectivity for the target. This spatial representation is called a *radar image*, and is a means for visualizing the target features responsible for scattering the transmitted electromagnetic radiation. Optical imagery is also a two-dimensional spatial representation of reflectivity. Unlike optical imagery, radar imagery may not provide a means

by which an object in a scene can be recognized by a typical human observer. However, radar imagery can be sufficient to characterize an object by its radar reflectivity alone. One method for creating such a radar image is called *direct imaging.* In a direct imaging system a pulsed range-gated radar with a narrow beam is scanned across a target while the intensity of the received signal is displayed as a function of spatial coordinates (Figure 1.2). Although this method is simple it has several disadvantages [4]:

a. In order to get high spatial resolution (sub-meter) the system must operate with a large bandwidth and have a very large aperture.
b. The cross-range resolution depends on the antenna beamwidth, which increases as the range increases, resulting in poor resolution.
c. The target may not exhibit the same scattering characteristics that it does when the entire target is illuminated at once.

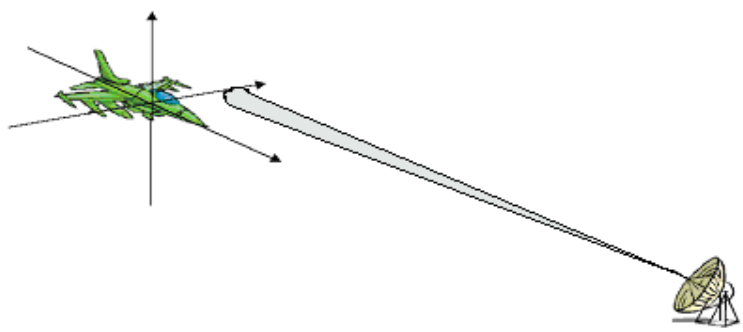


Figure 1.2 Direct Imaging Method

### 1.3.1. RESOLUTION

Radar image resolution is normally referred to as the ability to distinguish two closely spaced elements of a target. In radar imagery there are two dimensions of image resolution called range resolution and cross-range resolution. Basically, discrimination in range is

achieved by measuring the time delay between distinct features of the transmitted waveform and those same features in the received waveform. The range to the target is computed using the measured delay and knowledge of the propagation velocity of the transmitted radiation. Obviously, the most basic transmitted waveform can be a train of pulses as shown in Figure 1.3.

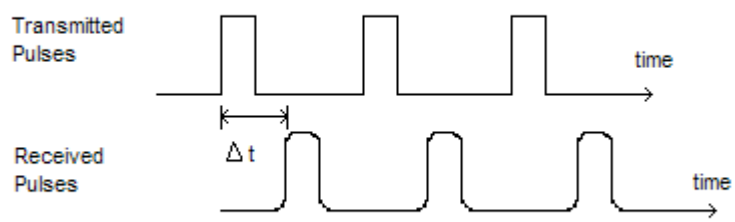



Figure 1.3 Pulse train for range measurement

In the case of a train of pulses the leading and trailing edge of a pulse provides a time marker. Range resolution is limited by the bandwidth of the radar system and can be simply expressed as

$$\rho_y = \frac{c}{2B_c} \ , \tag{1.1}$$

where c is the speed of propagation of the electromagnetic radiation and $B_c$ is the radar bandwidth. The radar bandwidth is actually the receiver subsystem bandwidth, or the point at which the frequency response drops to half its maximum ("3dB bandwidth"). The 3dB bandwidth is related to the pulse width by

$$B_c = \frac{0.886}{\tau} \approx \frac{1}{\tau} \ . \tag{1.2}$$

In a pulsed radar system the pulse duration required to get high resolution in range is very short. However, the use of a very short pulse makes it impossible to deliver adequate energy per pulse to the target needed to produce a sufficient return signal to noise ratio (SNR) for detection. In order to overcome this limitation, wide bandwidth waveforms such as chirped or stepped frequency waveforms are often employed with pulse compression techniques. This allows the radar system to achieve longer pulse durations and the higher SNR needed for high-resolution radar imaging. A complete treatment of these waveforms and the pulse compression techniques used for high-resolution radar can be found in several good radar texts [3, 4, 10].

Analogous to an optical imaging system, the cross range resolution of a direct imaging radar system depends on the antenna aperture size. In an optical imaging system free from aberrations the *limiting resolution*, or center-to-center separation of two points in the image that can be resolved, is defined as

$$\Delta s \approx f\lambda / D \ , \tag{1.3}$$

where D is the aperture diameter, $f$ is the imaging system focal length and $\lambda$ is the center wavelength of the electromagnetic radiation passed through the system [14]. It is clear that in order to get better resolution the aperture must get larger and the wavelength shorter. Since radar systems use wavelengths several orders of magnitude larger than optical systems, it would be necessary to use extremely large apertures with a direct imaging radar system in order to approach the spatial resolution capabilities of an optical imaging system.

As stated above, cross-range resolution for direct imaging radar is mainly determined by the antenna beam width, which is a function of the antenna aperture size. Two targets separated by an amount $\Delta s$ at a distance $R$ from the radar antenna, can be resolved only if

they are not both in the radar beam at the same time. Therefore, the required separation between two point-like targets must satisfy the expression

$$\Delta s = R\lambda / L \ , \tag{1.4}$$

where $R$ is the slant range from the radar antenna to the target, $\lambda$ is the radar center wavelength and $L$ is the diameter of the radar antenna (aperture size). Again, one main disadvantage of this method is that the resolution is a function of range.

The resolution disadvantages of the direct radar imaging method can be overcome by *synthetic aperture imaging* methods. A synthetic aperture radar (SAR) image is the result when several observations of an object at different frequencies and angles are coherently combined. Figure 1.4 shows the geometry of a spotlight mode SAR imaging scenario. Each individual observation of the target at a particular angle with respect to the radar flight direction makes up a sequence of narrow frequency band, single angle measurements of the target reflectivity. The process of coherently combining these individual observations allows the SAR system to achieve spatial resolutions comparable to direct imaging radar systems that might employ large bandwidths and very large apertures. The range resolution $\rho_y$ of synthetic aperture radar is also determined by the bandwidth of the radar (Equation 1.1), and the cross range resolution is determined by the angular diversity that results from a synthesized angular aperture. The cross range resolution for a SAR collection can be calculated by

$$\rho_x = \frac{\lambda}{2\Delta\theta} . \tag{1.5}$$

where $\lambda$ is the center wavelength of the radar and $\Delta\theta$ is the angular aperture extent. The factor of two is due to the fact that the phase differences between elements of the synthetic aperture result from a two-way path difference and are, therefore, two times larger than in the

case of a real aperture. If a full synthetic aperture is formed, the azimuth spatial resolution is completely independent of range and is determined only by the synthetic aperture size. Since it is desirable with a SAR system to illuminate the entire imaged scene it is also necessary to use an antenna small enough to provide a wide beamwidth. Although it seems that the smaller the SAR antenna the wider the beamwidth and therefore, the better the resolution, there is still an issue with providing adequate energy to the target at large distances in order to have a good SNR. Thus, a sufficiently long antenna is required in order to focus adequate power.

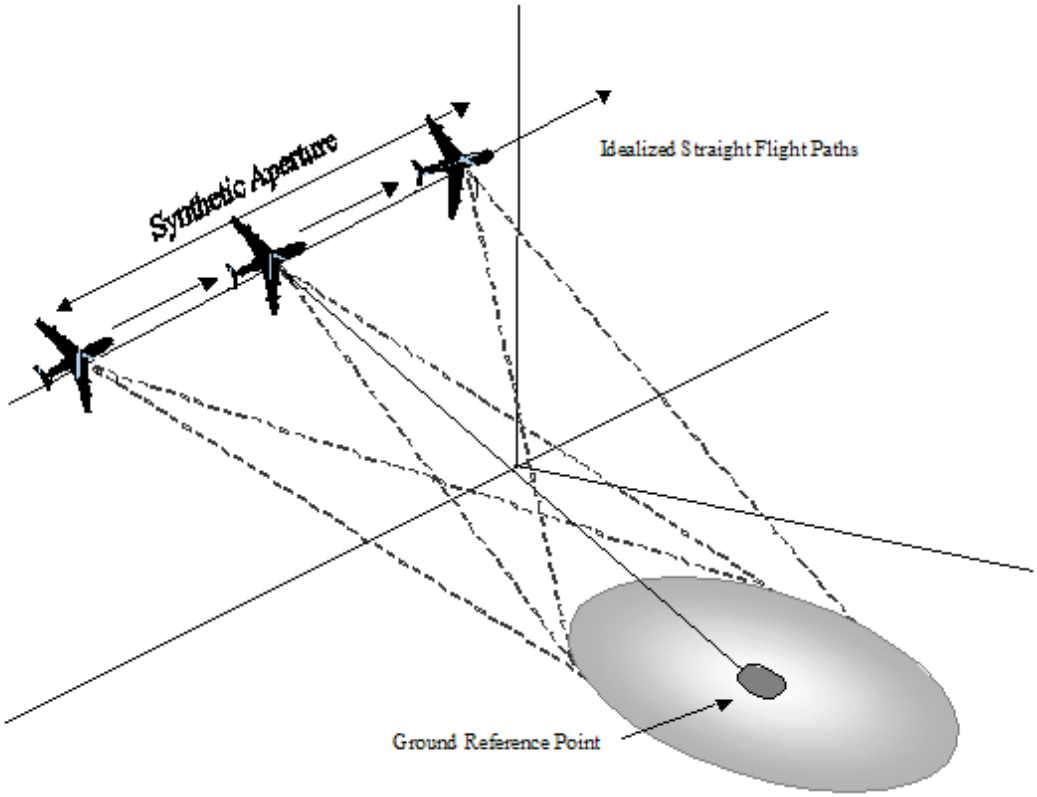



Figure 1.4 Synthetic aperture radar geometry

The mode of synthetic aperture radar imaging that is explored in the rest of this thesis is Inverse Synthetic Aperture Radar (ISAR) imaging. ISAR imaging is similar to spotlight SAR. However, instead of the radar platform moving around the target scene as in Figure 1.4,

with ISAR the radar platform is stationary and the target undergoes motion. ISAR imaging is a powerful tool often used for advancing the discipline of Automatic Target Recognition (ATR) and Identification. Since ISAR involves actual target motion in order to create a synthetic aperture it is often employed for imaging of ships, ballistic missiles, and in-flight aircraft.

The imagery processed and presented in this thesis originated from both simulated and collected ISAR data sets. In the following chapters the fundamental concepts behind ISAR imaging will be presented along with several signal processing approaches to image formation. Several methods of forming imagery from both simulated and real ISAR data will be demonstrated including the matched filter algorithm, FFT reconstruction, and high-resolution spectral estimation algorithms. A qualitative and quantitative comparison between these methods and algorithms will be made to determine their usefulness in extracting information from ISAR imagery.

## 1.4 PURPOSE AND DESCRIPTION OF RESEARCH

The purpose of this research was to study and assess the use of two-dimensional classical and modern spectral estimation techniques for enhancing inverse synthetic aperture radar imagery, and to develop a MATLAB toolbox for use with turntable and chamber ISAR data sets that allows a user to compute ISAR imagery using many 2-D algorithms methods presented in this thesis.

### 1.4.1. ORGANIZATION OF THESIS

This thesis is organized as follows:

CHAPTER 2 is an overview of inverse synthetic aperture radar and includes a description of the turntable ISAR data collection environment and geometry. The two-dimensional matched filter algorithm for image formation is also introduced and applied to both simulated and real ISAR data. The topic of zero-Doppler clutter and a method for mitigating its effect on imagery is introduced. The geometry of the data collection necessitates an introduction to methods for polar-to-cartesian conversion in order to use Fourier based image formation techniques. Several methods of this conversion will be discussed. Using simplifying assumptions the two-dimensional Discrete Fourier Transform is derived from the 2-D matched filter algorithm and used to compute imagery from real experimental ISAR data from a turntable range.

CHAPTER 3 introduces several two-dimensional spectral estimation techniques. Both classical and modern spectral estimators are introduced and used as image formation algorithms for the simulated and real data sets. Imagery is computed and presented for qualitative comparison.

CHAPTER 4 summarizes the completed research, quantitatively compares the results of the presented image formation algorithms, outlines all conclusions made, and offers suggestions for future work.

## 1.5 RESEARCH CONTRIBUTIONS

The primary contribution of this work is the thorough analysis of several classical and modern spectral estimation methods to compute enhanced imagery from both simulated and experimental ISAR data. Tradeoffs between these image formation methods are studied and presented in this thesis. The steps required to produce well focused imagery using data collected from a real ISAR collection system are outlined, and a ISAR processing toolbox is created using MATLAB that can be used with turntable and chamber data sets to create complete ISAR imagery. The usefulness of each image formation algorithm for applications such as Automatic Target Detection (ATD) and automatic target recognition (ATR) is explored in the context of quantifying the resolution enhancement, variance and dynamic range of point-like scatterers and point-like target features.

# CHAPTER TWO

## 2.1 ISAR DATA COLLECTION

Inverse Synthetic Aperture Radar imaging differs from Synthetic Aperture Radar imaging by the fact that relative motion between the radar transmit/receive platform and the target is due to target movement instead of radar platform movement. Situations where this is possible may include: imaging a ship at sea that is turning or rocking in the waves [14], radar imaging of a ballistic missile, imaging of an aircraft in flight, and controlled turntable and chamber experiments. This research focused on turntable and chamber measurements and the processing of imagery from data collected from these systems.

The data processed for the first section of this thesis was collected at the Radar Cross Section (RCS) Turntable Range at the Georgia Tech Research Institute (GTRI). Actual pictures from the GTRI turntable range are shown in Figure 2.1.

Figure 2.1 (Left) GTRI RCS Turntable range (Right) T-72 Tank used for data ISAR data collection experiment

The diagram in Figure 2.2 diagrams the experimental configuration of the turntable collection setup at GTRI and outlines the radar collection geometry for a point target at coordinates $(x_{0,} y_0, z_0)$.

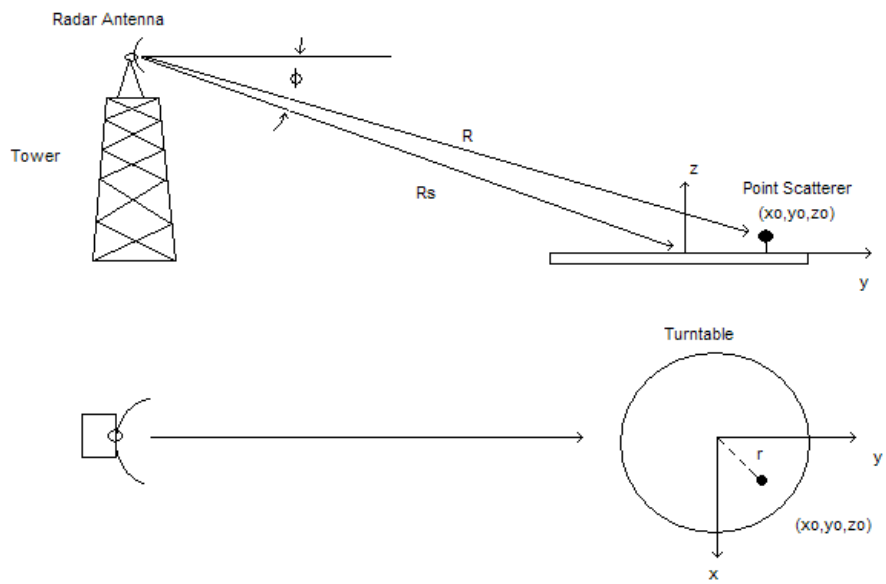



Figure 2.2 Turntable ISAR data collection setup and geometry

The GTRI radar transmits a stepped frequency waveform that consists of a series of L pulses. Each pulse has a sequentially higher carrier frequency than the last as shown in Figure 2.3.

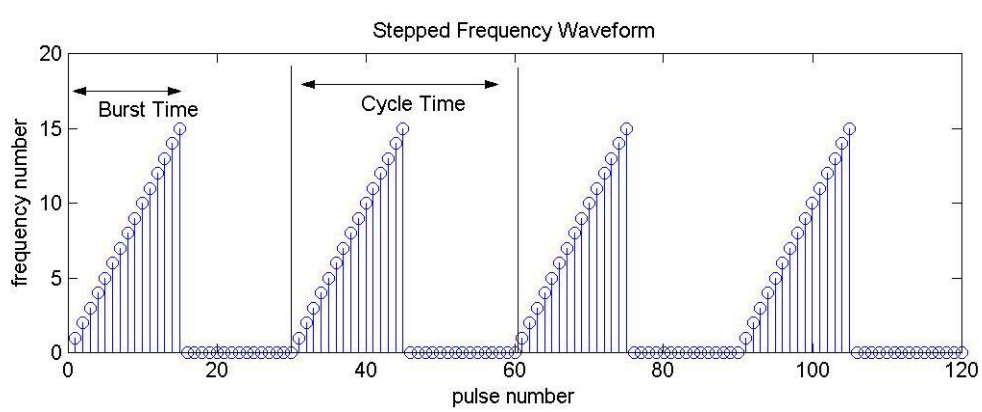



Figure 2.3 Example of stepped frequency waveform used at GTRI

In order to derive a method of image formation, it is convenient to imagine the radar element rotating in an arc around the target. The path of the radar element sweeps out an arc and forms the imaging aperture. The radar return bursts are sampled at equally spaced steps as the aperture is formed as shown in Figure 2.4. The samples are therefore collected on a polar grid. The distance from the radar element to the center of the turntable must be known. It is important to mention that this is a criterion for typical airborne SAR collection platforms that operate in stripmap mode or spotlight mode. In other words, the distance from the radar antenna to a known reference point in the imaged scene must be known with sub-wavelength accuracy. Phase errors often result from this assumption not being completely accurate. These phase errors must be corrected in order to compute well-focused imagery. The topic of correcting for focusing errors by utilizing methods such as *autofocus* is covered in detail in [13,16].

Using the setup diagram in Figure 2.2 the squared distance from any point on the aperture to a point target at $(x, y)$ can be expressed as

$$r^2 = R_s^{\,2} + d^2 - 2R_s d\cos(90 - \theta - \varphi) \ , \tag{2.1}$$

where $R_s$ is the slant range, $d$ is the distance from the turntable center point to the point target and $\varphi$ is the slant angle. After use of some trigonometric identities and the substitutions

$$\begin{aligned} d\sin\varphi &= y \\ d\cos\varphi &= x \end{aligned} \ , \tag{2.2}$$

an expression for the distance r from a point on the aperture to the point target $(x, y)$ is found to be

$$r(\theta, x, y) = R_s \left[ 1 + \frac{(x^2 + y^2)}{R_s^{\,2}} + \frac{2}{R_s}[y\cos\theta - x\sin\theta] \right]^{\frac{1}{2}} . \tag{2.3}$$

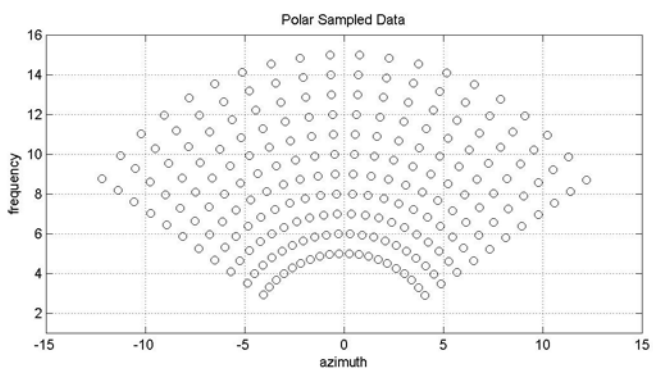



Figure 2.4 Imaging aperture or support showing data collection samples

The normalized field at the radar due to a point target at $(x, y)$ is

$$V(\theta, x, y) = \exp(-j2\pi f(2T_{delay})) \ , \tag{2.4}$$

where the distance r is used to used to compute the time delay $T_{delay}$ and $f$ is the center frequency of the radar. The factor of two is due to a two-way propagation of the signal. Substituting Equation 2.3 into 2.4, the normalized field due to a point target at $(x, y)$ can be expressed as

$$V(\theta, x, y) = \exp(-j\frac{4\pi}{\lambda}R_s\left[1+\frac{(x^2+y^2)}{R_s^{\ 2}}+\frac{2}{R_s}[y\cos\theta - x\sin\theta]\right]^{\frac{1}{2}}) \ . \tag{2.5}$$

## 2.2 2-D MATCHED FILTER

As mentioned in Chapter One, a synthetic aperture is formed by a coherent summation of all return pulses over a defined aperture. Therefore, the total response due to the point target at $(x, y)$ is the summation of the normalized field over the entire collection aperture.

$$G(x, y) = \sum_{\theta} V(\theta, x, y) \tag{2.6}$$

The total response of the synthetic aperture focused, or steered, to the point $(x, y)$ is

$$G(x, y) = \sum_{\theta} V(\theta, x, y)\exp(j\frac{4\pi}{\lambda}r(\theta, x, y)) \ . \tag{2.7}$$

This response calculated for each resolution cell in an image grid forms the basis for a geometric matched filtering algorithm to form focused images of the ISAR scene. The 2-D matched filter algorithm requires $MN^2 + 2MN + N$ computations, where $M$ and $N$ are the data record lengths for columns and rows respectively. The complexity of the matched filter algorithm lies in the fact that Equation 2.7 must be applied separately to each pixel, or grid location in the image. This method is a “brute force” technique that requires significantly more computational resources than other more efficient imaging techniques to be discussed in the following sections. However, the 2-D matched filter algorithm offers some advantages in certain applications where computing speed is not critical. It is also less restrictive to the collection geometry than other image formation methods. The results of applying a 2-D matched filter algorithm to the T72 tank ISAR data is shown in Figure 2.5.

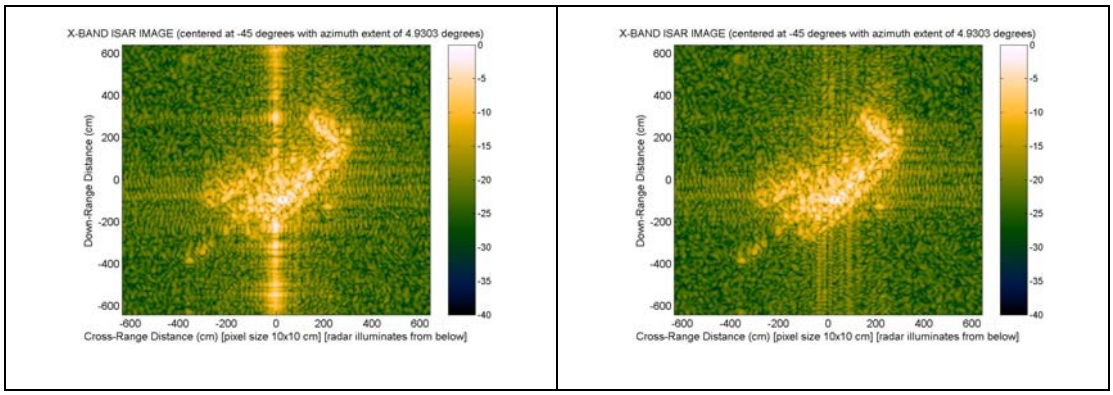



Figure 2.5 Two Results of matched filter image formation for ISAR data set from T72 Tank. No zero-Doppler clutter subtraction (Left), with zero-Doppler clutter subtraction (right).

It is worth noting that the geometric matched filter algorithm is often referred to as the "Back Projection Algorithm". It is one of the most common image formation algorithms used by satellite remote sensing ground systems such as the ones used for RADARSAT and the NASA Shuttle Radar Topography Mission (SRTM) [17].

## 2.3 ZERO DOPPLER ESTIMATION AND ZEROING

Returns from stationary clutter near the collection platform will negatively impact the radar image by introducing zero-Doppler clutter in the ISAR image [18, 19]. In the T-72 data collected at GTRI this clutter is primarily due to the grass field around the turntable and is expressed as a strong return line running through the image at zero cross-range (see left image in Figures 2.5). Since the entire target is uniformly illuminated during the collection process, side-lobe energy returns from stationary objects around the turntable will be imaged into the zero-Doppler bin after image formation.

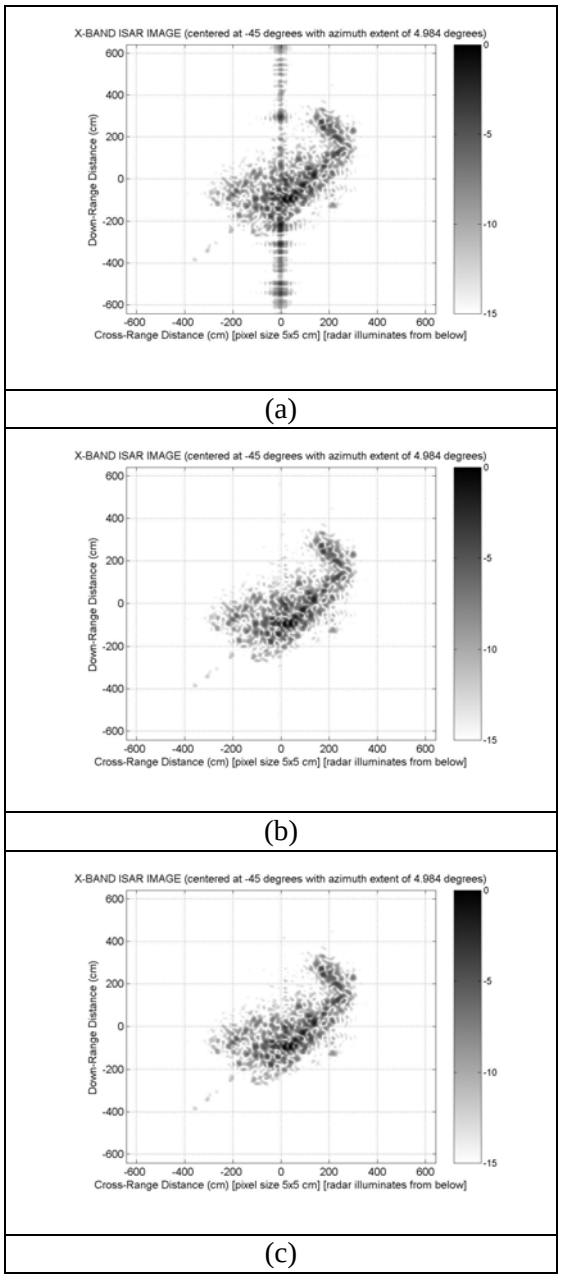



Figure 2.6 Results of ZDC subtraction and zeroing for (a) no ZDC subtraction (b) ZDC subtraction with 50 degree aperture (c) ZDC subtraction with 7 degree aperture.

One method for reducing the effect of this zero-Doppler clutter on the ISAR image is to find the mean value of each frequency bin over the entire 360° set of data. The resulting means are then subtracted from the image raw data prior to any processing or image

formation. The result of applying a ZDC subtraction process to the T-72 data is displayed in Figure 2.6. Even better results can be achieved by using an azimuth aperture for the ZDC subtraction that is only slightly larger than the azimuth extent used to form the ISAR image. Results from ZDC subtraction using azimuth apertures of 50 degrees and 7 degrees are shown in Figure 2.6b, and 2.6c respectively. This method is convenient since it can be applied to the raw data set, or phase history, thereby creating a new data set that is then used for image formation. Using azimuth apertures slightly larger than the azimuth extent used for the image formation gives the best results at the expense of some loss due to mixed pixels. Since the zero-Doppler clutter can decorrelate over time (angle), ZDC subtraction can be performed prior to each image formation as a set of images is created for the entire 360° rotation of the target.

## 2.4 POLAR TO RECTANGULAR INTERPOLATION

In order to use Fourier based methods for 2-D image formation the collected data must be sampled on a uniform rectangular grid. Due to the collection geometry of the turntable ISAR system the data is collected on a polar grid as shown in Figure 2.4. There are several methods for converting sampled data from a polar coordinate collection to a Cartesian grid [20,21]. This is often called *polar-to-Cartesian conversion* or *polar-to-Cartesian interpolation.* One of the most common methods is the use of a separable 1-D sinc algorithm. This algorithm is based on the motivation that a radial slice of the 2-D Fourier transform is a 1-D transform of a projection. Therefore, according to the sampling theorem a 2-D transform can be recovered from a radial slice from samples on the slice with spacing of $2\pi / L$ radians per unit of length. The algorithm proceeds as follows. A windowed 1-D sinc interpolation kernel is applied in the radial dimension to produce samples that are uniformly spaced in

frequency. Then a windowed sinc kernel is applied in the orthogonal direction to get samples that are uniformly spaced in azimuth, or time. The final result is a Cartesian grid of sampled data that is ready to be used with the 2-D Fourier transform.

Another method used in this work is a two-stage interpolation algorithm that first interpolates the complex data with a range-angle (R-theta) interpolation. Then the polar sample nearest to the rectangular grid sample location of interest is located. This is done with a nearest neighbor search in both the radial and azimuth index directions. This polar sample is then assigned to the nearest rectangular grid index.

In order to apply this conversion method to the phase history it is necessary to form an inscribed rectangular grid over the region of support. The coordinates for the inscribed square must be calculated from the aperture extent. An outline diagram of this polar-to-Cartesian conversion process in preparation for image formation is shown in Figure 2.7.

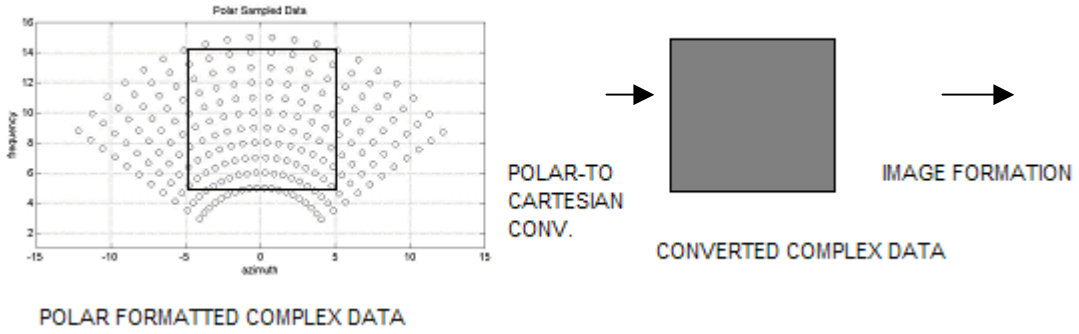



Figure 2.7 Polar-to-Cartesian interpolation prior to Fourier image formation.

Usually it is desirable to choose an inscribed rectangular aperture that will result in equal cross-range and down-range resolutions after image formation. This is possible by choosing the aperture dimensions according to the calculation in Appendix A.

## 2.5 2-D DISCRETE FOURIER TRANSFORM

Making a few assumptions about the ISAR geometry allows for the development of a computationally more efficient image formation method than the matched filter algorithm. The argument of the exponential of Equation 2.5 can be expanded in a Taylor series to be

$$r \approx R_s \left[ 1 + \frac{1}{R_s}[y\cos\theta - x\sin\theta] + \frac{(x^2 + y^2)}{2R_s{}^2} \right] \quad . \tag{2.8}$$

Using the following assumptions

$$\begin{aligned} &R_s \gg x, y \\ &\cos\theta \approx 1 \quad , \\ &\sin\theta \approx \theta \end{aligned} \tag{2.9}$$

the expression in Equation 2.8 can be further reduced to

$$r \approx R_s + y - x\theta \quad . \tag{2.10}$$

Using this in Equation 2.7 for the total response of the focused synthetic aperture steered to the point $(x, y)$, we are left with

$$G(x, y) = \sum_{\theta} V(\theta, x, y) \exp(j\frac{4\pi}{\lambda}(R_s + y - x\theta) \quad , \tag{2.11}$$

which further simplifies to

$$G(x, y) = \exp(-j\frac{4\pi}{\lambda}(R_s + y)) \sum_{\theta} V(\theta, x, y) \exp(-j\frac{4\pi}{\lambda} x\theta) \tag{2.12}$$

Taking the magnitude of this expression yields

$$\left|G(x, y)\right| = \left| \sum_{\theta} V(\theta, x, y) \exp(-j\frac{4\pi}{\lambda} x\theta) \right| \quad . \tag{2.13}$$

This is the magnitude of a DFT sum. Equation 2.13 indicates that Fourier transforming the field values that were sampled along the rectangular aperture will form an image. It should be

recognized that in order to arrive at Equation 2.13, approximations were made assuming large observation distances and small angles subtended by the aperture.

The advantage of the 2-D DFT is that it provides a method of image computation that is significantly less computationally intensive than the 2-D matched filter approach. For an array of dimension $MxN$, the 2-D DFT computed using the Fast Fourier Transform requires $MN\log_2(N) + NM\log_2(M) + MN$ computations. This is a significant increase in efficiency over the 2-D matched filter algorithm. Imagery results from the 2-D DFT approach utilizing the FFT for computational speed, and using various windowing techniques to reduce sidelobes, are shown in the Figure 2.8.

It can be seen in Figure 2.8, that any application of windowing in order to reduce sidelobes will also reduce the resolution in the imagery. Several weighting (windowing) functions can be used to reduce sidelobes, including the Hamming, Nuttall, Hanning, and Taylor. The Taylor and Hanning weighting functions are popular in SAR imaging applications. There also exist spatially variant windowing techniques (SVA) that have been shown to achieve reduced sidelobe performance with increased mainlobe resolution [38]. These methods have been shown to be useful for large area SAR imagery, where the imaged scene may contain many targets or features of interest. Although useful in some applications, SVA and related methods will not be explored in this thesis.

Imagery formed with the 2-D DFT approach exhibit speckle that is due to the coherent nature of the SAR data collection and the fact that the DFT method does not treat the signal history statistically, i.e., it does not average the across resolution cells. Reduction of speckle can be achieved without loss of resolution when modern spectral estimation methods are used to compute imagery [38].

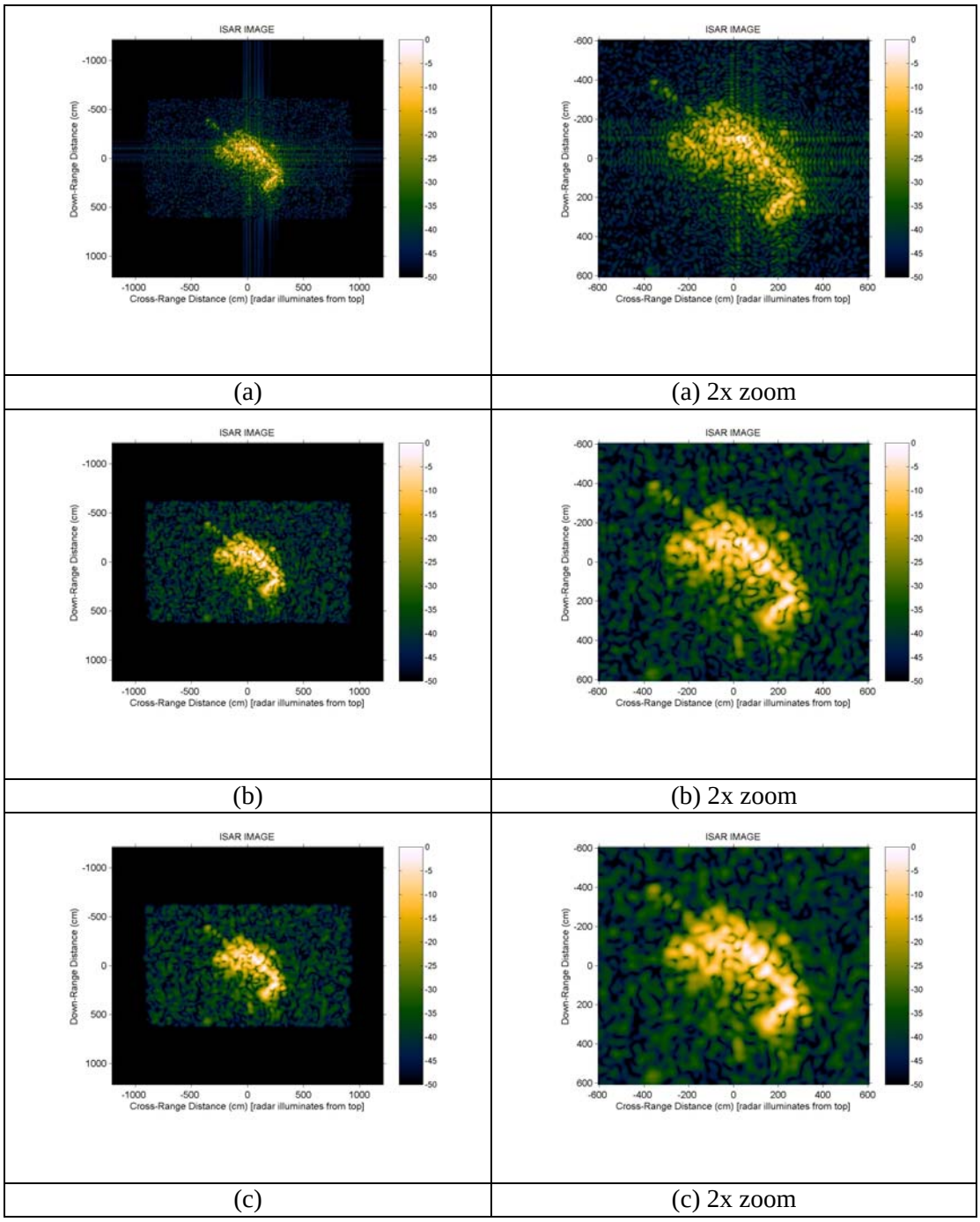



Figure 2.8 Results of FFT image formation (a) rectangular window (b) Hamming window (c) Nuttall window

# CHAPTER THREE

## 3.1 TWO-DIMENSIONAL SPECTRAL ESTIMATION

In this section classical and modern two-dimensional spectral estimation techniques will be presented and applied to computing simulated and real ISAR imagery. In a typical ISAR system, high resolution in range must be obtained through the use of wide bandwidth transmission, and high cross range resolution must be obtained by coherent integration of the transmitted radar pulses over the synthetic aperture. Imagery is commonly obtained through conventional Fourier processing. However, in many radar systems only a limited bandwidth is available and ideal synthetic aperture extents are not possible, resulting in a finite data record of limited size. In general the resolution of Fourier techniques are limited by the length or aperture of the data record. Due to the artificial truncation of the frequency spectrum at the radar bandwidth, and the azimuthal aperture, ISAR imagery can suffer from sidelobe signatures that can be confused for target features or limit the use of the ISAR imagery. Although sidelobes can be reduced by an appropriate windowing function, sidelobe reduction comes at the price of further reduction of overall resolution, as was shown in Figure 2.8. There has been research in the development of alternative techniques for ISAR image formation that provide methods of overcoming the inherent resolution limitations of typical radar systems and Fourier techniques. The formation of ISAR imagery can be viewed as a spectral estimation problem, and therefore, both classical and modern spectral estimation methods can be used to compute ISAR imagery. Many modern spectral estimation methods are based on parametric modeling of a finite signal and can be used to achieve increased resolution beyond the capabilities of Fourier based techniques. In the following section, several two-dimensional classical and modern spectral estimation methods will be presented and applied to simulated and collected ISAR data, and the results compared to those achieved

through the application of the 2-D FFT or Matched Filter image formation methods introduced in chapter two.

## 3.2 2-D CLASSICAL SPECTRAL ESTIMATION METHODS

The two main classical spectral estimation methods presented in the following section are the Welch periodogram and the Blackman-Tukey correlogram. Although both of these methods utilize the Fourier transform, the method by which they are computed differs, and the results when used to compute ISAR imagery have different outcomes.

### 3.2.1 2-D PERIODOGRAM

A *periodogram* is a PSD estimate based on a direct transformation of a finite data set followed by an averaging process. The periodogram seeks to provide a statistically stable and smooth spectral estimate from the data record. A related PSD estimate called the *correlogram* is based on first forming the correlation estimates from the data and then performing a Fourier transform on the estimates. When dealing with finite length discrete time sampled sequences there are many tradeoffs to consider in order to produce statistically reliable spectral estimates with a desired resolution. Some of these tradeoffs are data window type, time-domain or frequency-domain averaging, and ensemble averaging. A good treatment of these issues can be found in [22].

The diagram in Figure 3.1 outlines the Welch periodogram method for a 1-dimensional signal. The signal is first partitioned into smaller signal sections or sub-apertures. These subapertures may be selected to overlap successively as shown. An appropriate window is applied to each subaperture prior to transformation and estimation of the power

spectral density (PSD). The PSD estimates from each subaperture process are averaged to give the final periodogram PSD.

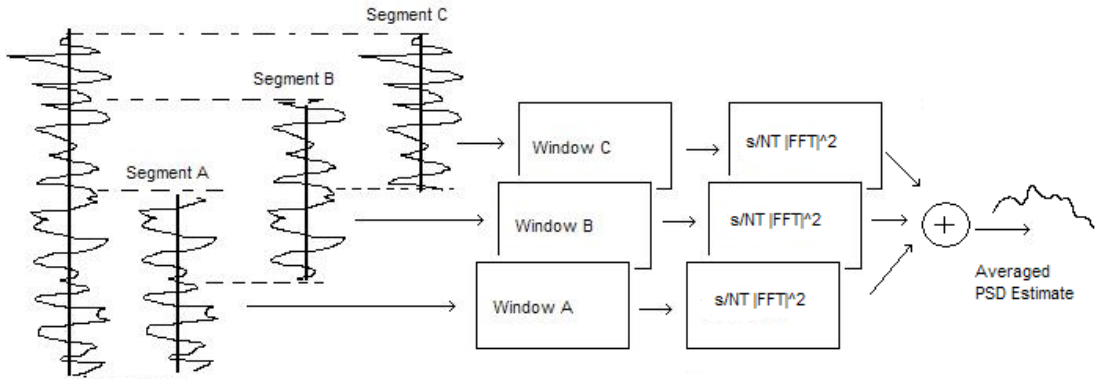



Figure 3.1 Outline for 1-D Welch periodogram method

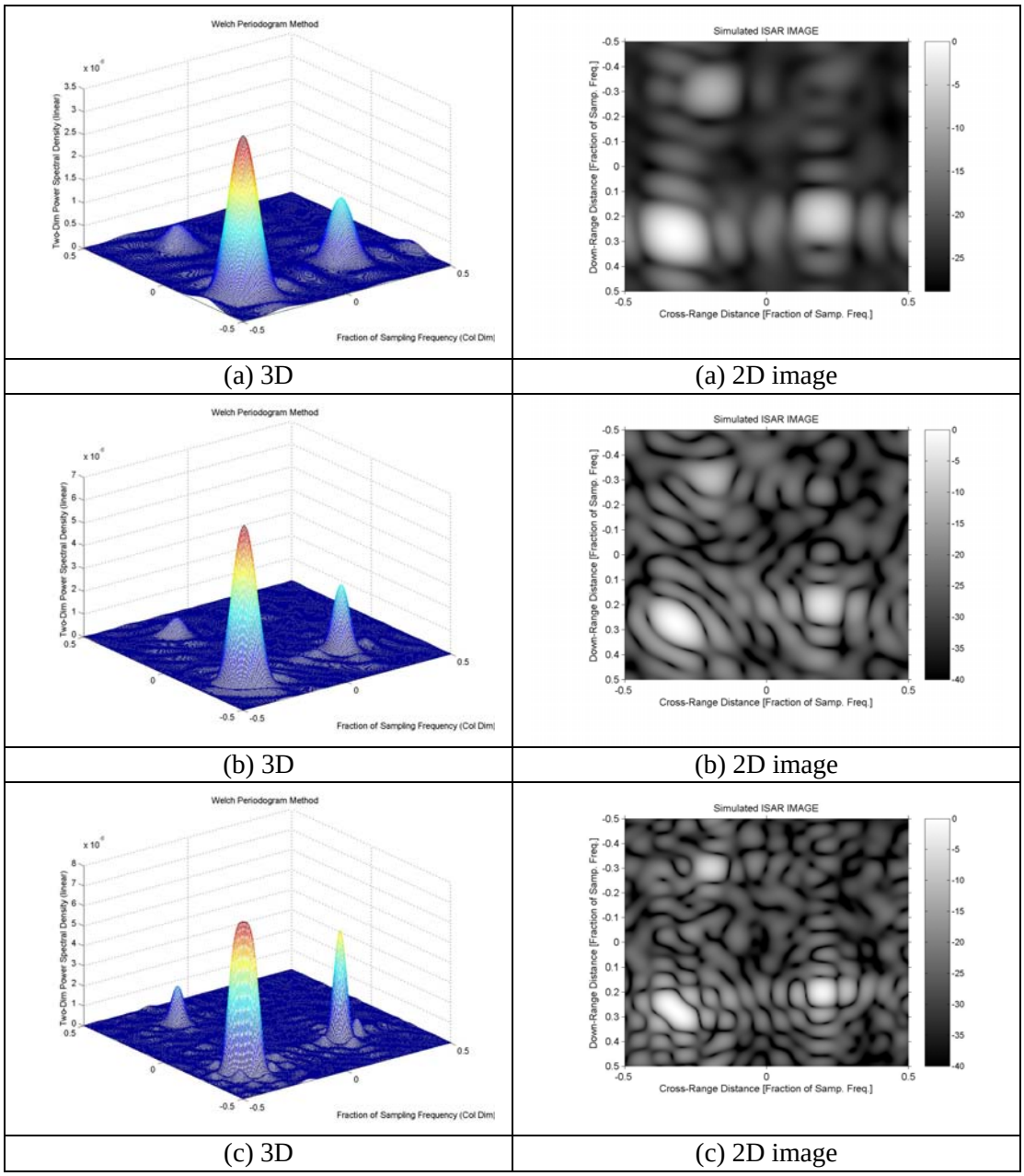
The two-dimensional Welch periodogram method can be understood by extending the 1-D method to 2-D subarrays and is mathematically represented by the following expression:

$$\hat{P}_{per}(f_1, f_2) = \frac{1}{MN}\left|\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} w[m,n]x[m,n]\exp(-j2\pi[f_1 m + f_2 n])\right|^2 , \qquad (3.1)$$

where $M$ and $N$ are the row and column lengths of the 2-D phase history data, $w[m,n]$ is an optional windowing function, and $x[m,n]$ is the 2-D phase history data, and $f_1$ and $f_2$ are the spatial domain samples in the image domain. In radar imaging applications, the periodogram method seeks to estimate the average power of the Fourier transform considering the stochastic nature of the clutter data. However, the subarrays can also be considered subapertures, and therefore will reduce the overall resolution because the subapertures are smaller than the full aperture, or data record.

When applied to ISAR raw data, the Welch method is similar to the 2-D Fourier Transform for image formation, but can result in a comparable loss of resolution due to the subaperture averaging. Figure 3.2 shows the results of the Welch periodogram image

formation algorithm for simulated ISAR data consisting of five point targets. It is easy to see that any subaperture formation and averaging reduces the resolution.




| | |
|---|---|
| (a) 3D | (a) 2D image |
| (b) 3D | (b) 2D image |
| (c) 3D | (c) 2D image |

Figure 3.2 Simulated ISAR images of five point targets with additive white noise from 16x16 data sample set. (a) Welch periodogram method with sub-array sizes of 8 for both columns and rows with 7 sample overlap (b) Welch periodogram method with sub-array sizes of 12 for both columns and rows with 4 sample overlap; (c) Welch periodogram method with sub-array sizes of 16 for both columns and rows and no overlap of sub-arrays.

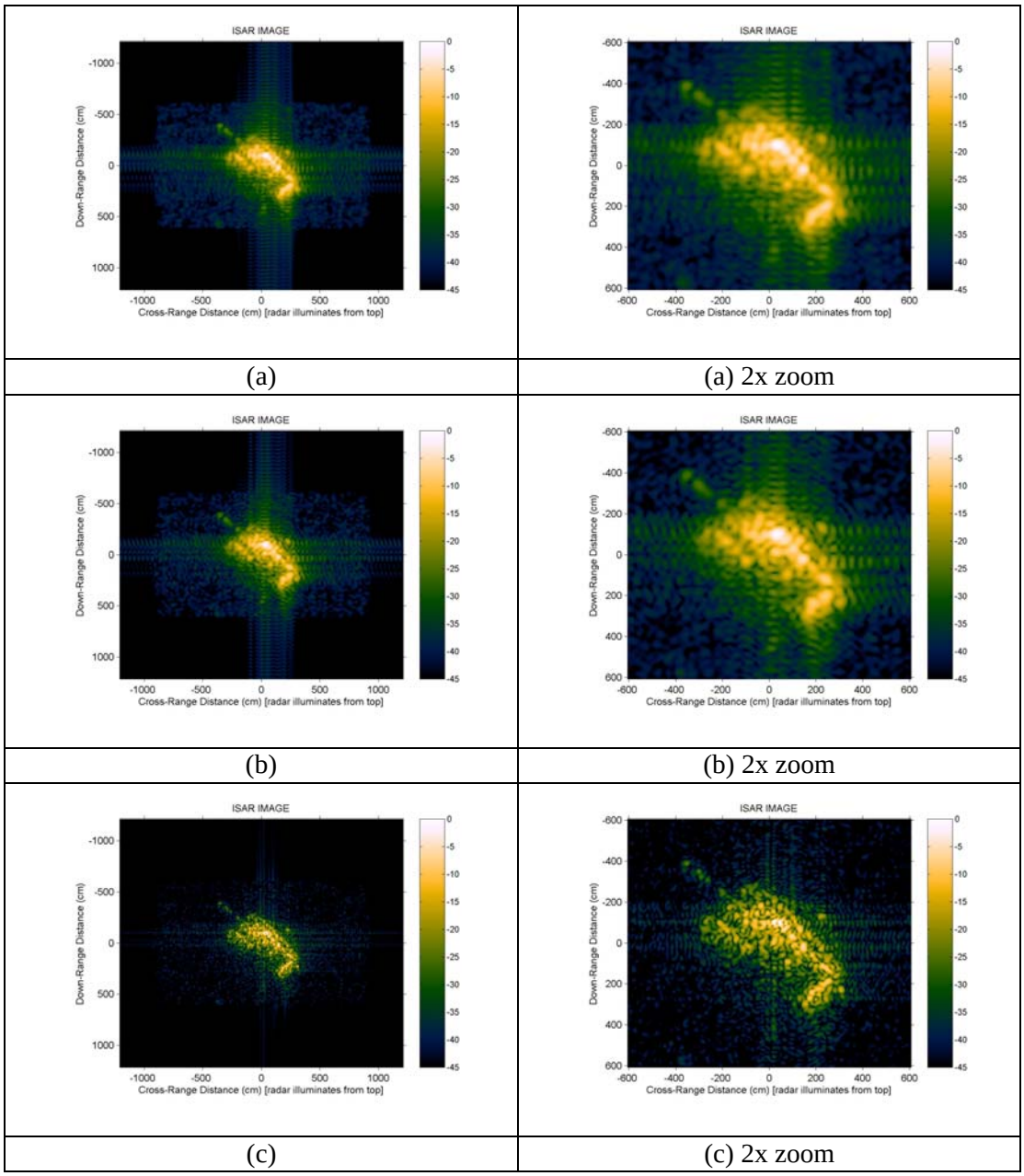



Figure 3.3 ISAR images of T-72 tank. (a) Welch periodogram method with sub-array sizes of 64 samples for both columns and rows and 32 sample overlap (b) Welch periodogram method with sub-array sizes of 64 samples for both columns and rows with no sample overlap (c) Welch periodogram method with sub-array sizes of 128 samples for both columns and rows and no overlap.

Even using the full data record the Welch periodogram is unable to resolve all five targets. It is also difficult to distinguish if the target on the far right is in reality two targets or one since the target of smaller amplitude is disguised within a side-lobe of the stronger target.

Results of the Welch periodogram method applied to the T-72 Tank data set are shown in Figure 3.3. Again, use of subaperture partitions and averaging blurs the imagery and results in loss of resolution.

### 3.2.2 2-D BLACKMAN-TUKEY

The Blackman-Tukey spectral estimate is a correlogram method that first requires the computation of the autocorrelation estimate from the signal data. The Blackman-Tukey spectral estimate was especially popular for analysis of 1-D signals prior to the broad availability of fast digital computers. Like the Welch periodogram method, the Blackman-Tukey method inherently involves a trade-off between smoothness (variance) and resolution. This trade-off is made by the selection of a maximum lag value when estimating the autocorrelation sequence. As the number of estimated lags increase, the resolution increases and the smoothness decreases. The Blackman-Tukey spectral estimate for two-dimensional signal data can be represented mathematically as

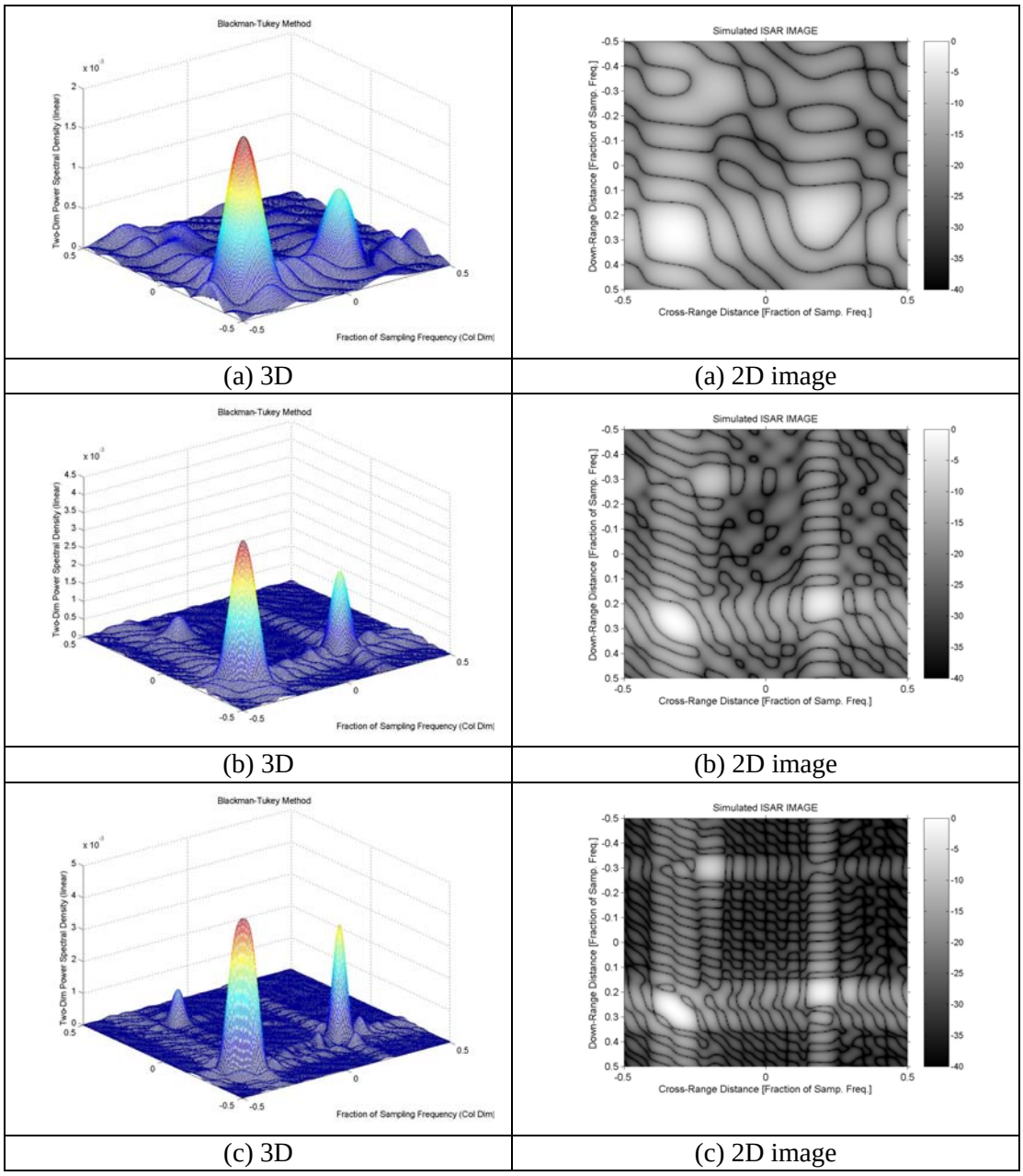
$$\hat{P}_{BT}(f_1, f_2) = \sum_{k=-p_1}^{p_1} \sum_{l=-p_2}^{p_2} w[k,l]\hat{r}_{xx}[k,l]\exp(-j2\pi[f_1 k + f_2 l]) \ , \tag{3.2}$$

where the biased autocorrelation sequence estimate (ACS) can be computed by

$$\hat{r}_{xx}[k,l] = \begin{cases} \dfrac{1}{MN}\displaystyle\sum_{m=0}^{M-1-k}\sum_{n=0}^{N-1-l} x[m+k,n+l]x^*[m,n], & for\left\{k \geq 0, l \geq 0\right\} \\ \dfrac{1}{MN}\displaystyle\sum_{m=0}^{M-1-k}\sum_{n=-l}^{N-1} x[m+k,n+l]x^*[m,n], & for\left\{k \geq 0, l < 0\right\} \\ \hat{r}_{xx}^*[-k,-l], & for\left\{k \leq 0, all(l)\right\} \end{cases} . \quad (3.3)$$

The imagery in Figure 3.4 shows the results from applying the Blackman-Tukey method to the simulated ISAR data from five point targets. The above stated tradeoff from selection of maximum lag is evident. The resolution is best in Figure 3.4c, which is computed from the ACS with the largest maximum lag. Figure 3.4a was computed from the smallest maximum lag of the three images and therefore has the poorest resolution. Even with a large maximum lag, the Blackman-Tukey method is unable to resolve the five point targets. The Blackman-Tukey method also suffers from higher amplitude sidelobes than the periodogram method, which may necessitate use of appropriate weighting functions or windows for sidelobe suppression.

Windowing of the ACS will further decrease the resolution of the spectral estimate with a tradeoff to suppress the sidelobes. The strong sidelobes are visible in the imagery and necessitate the use of appropriate windowing techniques for suppression. However, for comparison purposes no windowing is used here. The imagery in Figure 3.5c has the highest resolution due to having the largest maximum lag for the ACS estimate. However, the imagery in Figures 3.5a and b does a good job at isolating the strong scattering center near the center of the tank and may be useful for automatic target recognition applications where higher resolution is not necessary.




| | |
|---|---|
| (a) 3D | (a) 2D image |
| (b) 3D | (b) 2D image |
| (c) 3D | (c) 2D image |

Figure 3.4 Simulated ISAR images of five point targets with additive white noise, 16x16 data sample set. (a) AR Blackman-Tukey method with 4 ACS estimates (b) AR Blackman-Tukey method with 8 ACS estimates (c) AR Blackman-Tukey method with 12 ACS estimates. Results of the Blackman-Tukey method applied to the T-72 Tank data set are shown in figure.

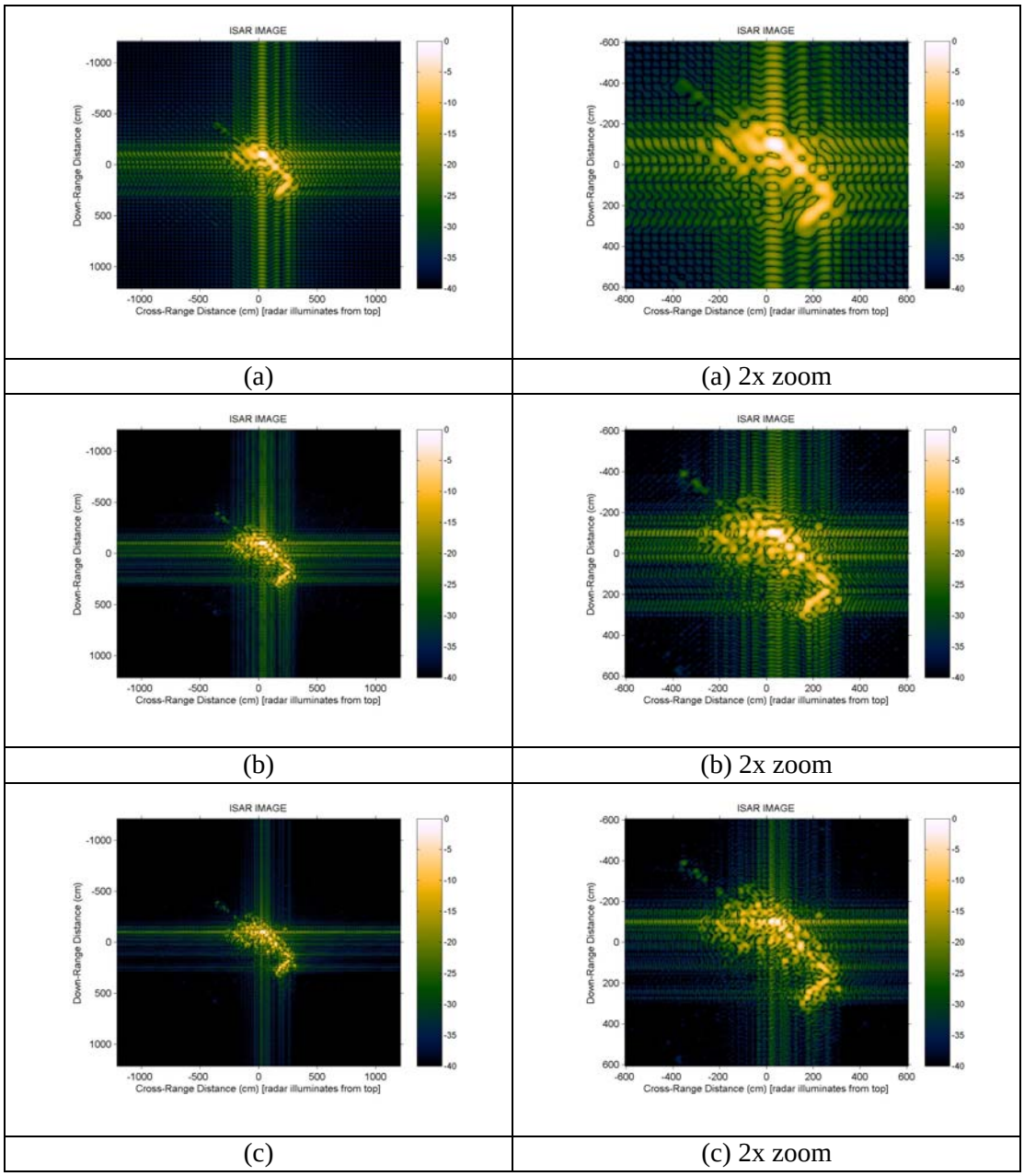



Figure 3.5 ISAR images of T-72 tank. (a) Blackman-Tukey method with 32 ACS estimates (b) Blackman-Tukey method with 64 ACS estimates (c) Blackman-Tukey method with 96 ACS estimates.

### 3.2.3 2-D MODERN SPECTRAL ESTIMATION METHODS

Modern spectral estimation techniques are model-based techniques that provide powerful methods for achieving higher resolution in ISAR imagery. Resolutions beyond the capabilities of Fourier techniques can be achieved in both range and cross-range by an order of two or more and data sets that have been corrupted and are otherwise of little use for imaging can be repaired to yield high-quality images [23]. Two-dimensional autoregressive spectral estimators have received the most attention in the literature and are the result of parametric models of 2-D random processes. Several autoregressive (AR) methods will be presented in the following sections. These methods vary by their computational approach and restrictions. Results of each method applied to simulated and real ISAR imagery is presented and compared. A 2-D minimum variance method will also be introduced.

### 3.2.4 2-D AR YULE-WALKER METHOD

Driving a 2-D linear shift invariant filter with a 2-D white noise process $w[m,n]$ will generate the two-dimensional autoregressive sequence

$$x[i,j] = -\sum_m \sum_n a[m,n]x[i-m, j-n] + w[i,j] \tag{3.4}$$

in which the region of support determines the range for the indices $m$ and $n$ [22]. Multiplying Equation 3.4 by $x^*[m-k, n-l]$ and taking the expectation will yield

$$\sum_i \sum_j a[i,j] r_{xx}[k-i, l-j] = \begin{cases} \rho_w & for\ [k,l]=[0,0] \\ 0 & otherwise \end{cases}, \tag{3.5}$$

which is the 2-D Yule Walker equations for a causal 2-D AR process. The 2-D Yule Walker equations provide a set of linear equations that relates the 2-D AR parameters to the 2-D ACS.

These equations for the quarter plane support region can be arranged into a block matrix form and expressed simply as

$$\underline{R^r}\,\underline{a_1^r} = \underline{\rho_1^r} \tag{3.6}$$

where the row ordered block matrix $\underline{R^r}$ is

$$\underline{R^r} = \begin{pmatrix} R^r[0] & R^r[p_1] & \cdots & R^r[p_1] \\ R^r[p_1] & R^r[0] & \cdots & R^r[p_1-1] \\ \vdots & \vdots & \ddots & \vdots \\ R^r[-p_1] & R^r[-p_1+1] & \cdots & R^r[0] \end{pmatrix}. \tag{3.7}$$

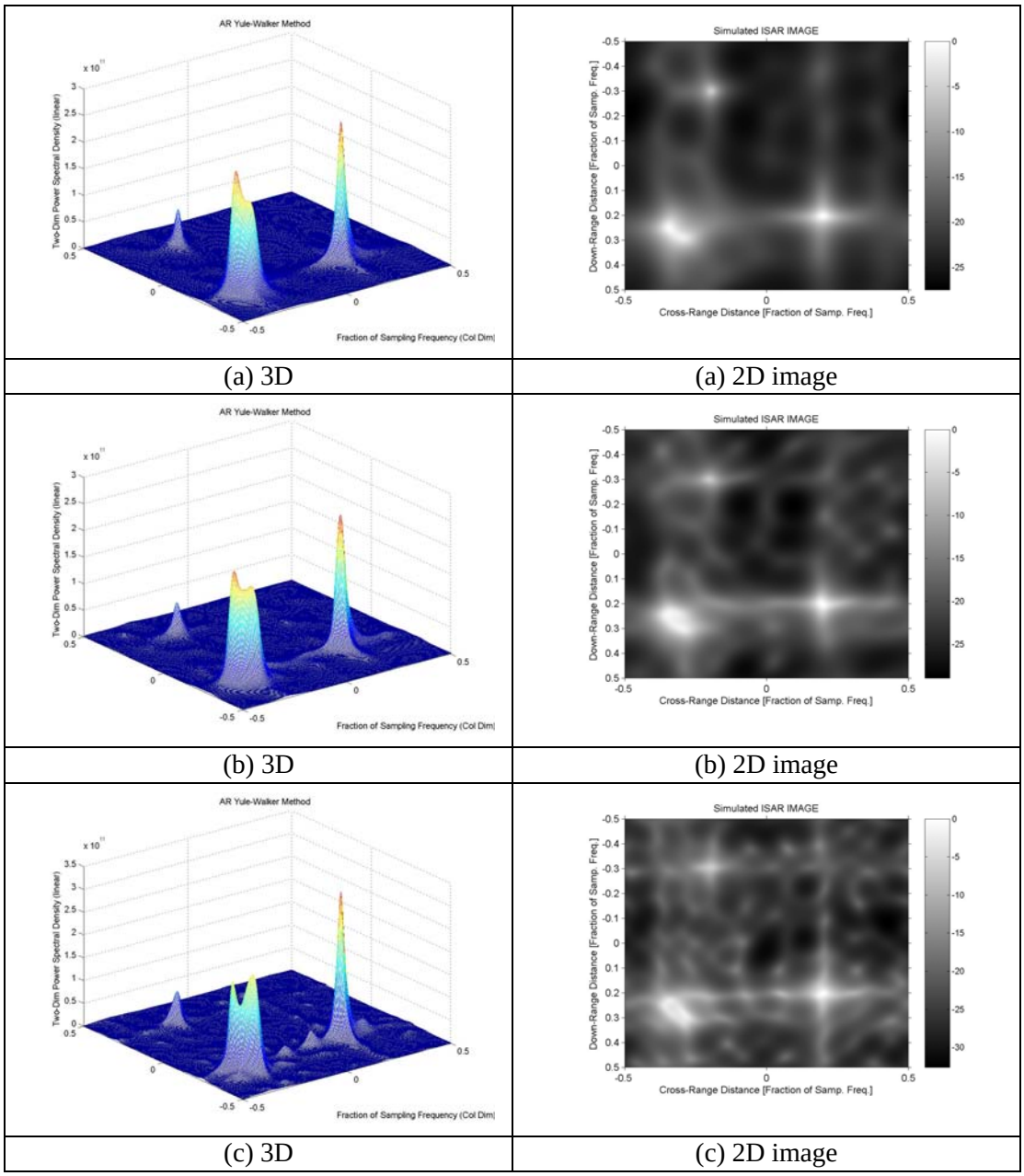
Each matrix element is also an $m\,x\,n$ ACS matrix that is Toeplitz in structure. The block matrix $\underline{R^r}$ is therefore *doubly Toeplitz* and can be exploited to develop a fast computational algorithm for the solution of the 2-D Yule Walker equations. The solution utilizes the *multichannel Levinson algorithm* to find the AR parameter estimates. A detailed development of the 2-D AR Yule Walker equations for all regions of support, and a discussion of a multichannel Levinson algorithm for solving the 2-D Quarter-Plane Yule-Walker equations can be found in [22]. The 2-D AR power spectral density is computed by

$$P_{AR}(f_1, f_2) = \frac{\rho_w}{\left|\sum_{m=0}^{p_1}\sum_{n=0}^{p_2} \hat{a}_1[m,n]\exp(-j2\pi[f_1 m + f_2 n])\right|^2}, \tag{3.8}$$

where $\rho_w$ is the 2-D white noise variance and $\hat{a}_1[m,n]$ is an array of autoregressive parameters. Usually a PSD for combined first and third quarter planes of support is computed, as described in [24].

The results of the AR Yule-Walker method used to compute imagery from simulated ISAR phase history is shown in Figure 3.6. Higher resolution is achieved as compared to the classical spectral estimation methods, but the Yule Walker method is not able to resolve all

five point targets. A high model order also introduces small artifacts into the imagery as the model tries to fit the noise with peaks.




| | |
|---|---|
| (a) 3D | (a) 2D image |
| (b) 3D | (b) 2D image |
| (c) 3D | (c) 2D image |

Figure 3.6 Simulated ISAR images of five point targets with additive white noise, 16x16 data sample set. AR Yule Walker method with model order (p,q) for both columns and rows (a) AR(5,5) (b) AR(7,7) (c) AR(10,10).

The Yule-Walker method applied to imaging the T-72 tank is shown in Figure 3.7 for three different model order selections. It can also be seen in these images that high model orders can produce distortion in the imagery. Again, this is the autoregressive model attempting to fit the broadly distributed clutter using sharp localized peaks.

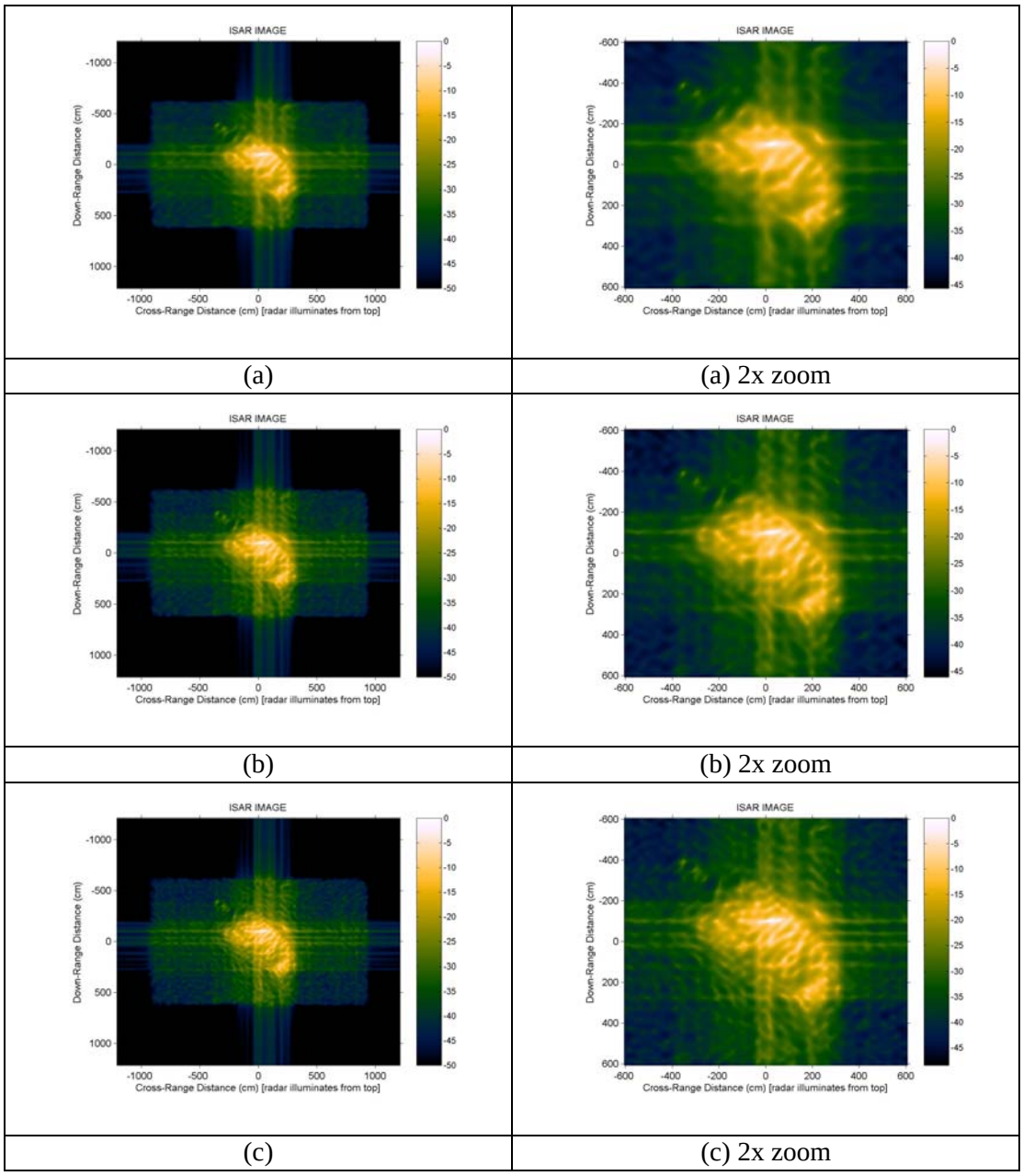



Figure 3.7 ISAR images of T-72 tank. AR Yule Walker method with model order (p,q) for both columns and rows (a) AR(30,30) (b) AR(40,40) (c) AR(50,50).

### 3.2.5 2-D AR LATTICE METHOD

The two-dimensional lattice method used for the following analysis is an extension of the one-dimensional *Burg* algorithm developed in the 1960s. The one-dimensional Burg algorithm, also called the "Maximum Entropy Method" (MEM), involves finding an all-pole model for the available data. First the reflection coefficients are sequentially computed by minimizing the sum of the forward and backward linear prediction squared errors.

$$\rho_{\mathrm{m}} = \frac{1}{2N} \sum_{n=m+1}^{N} \left( \left| e_m^f[n] \right|^2 + \left| e_m^b[n] \right|^2 \right) \tag{3.9}$$

$$\frac{\partial \rho_m}{\partial \operatorname{Re}\{k_m\}} + j \frac{\partial \rho_m}{\partial \operatorname{Im}\{k_m\}} = 0 \quad , \tag{3.10}$$

where the following recursive relationships are used in Equation 3.9

$$\begin{aligned} e_m^f[n] &= e_{m-1}^f[n] + k_m e_{m-1}^b[n-1] \\ e_m^b[n] &= e_{m-1}^b[n-1] + k_m^* e_{m-1}^f[n] \end{aligned} \quad . \tag{3.11}$$

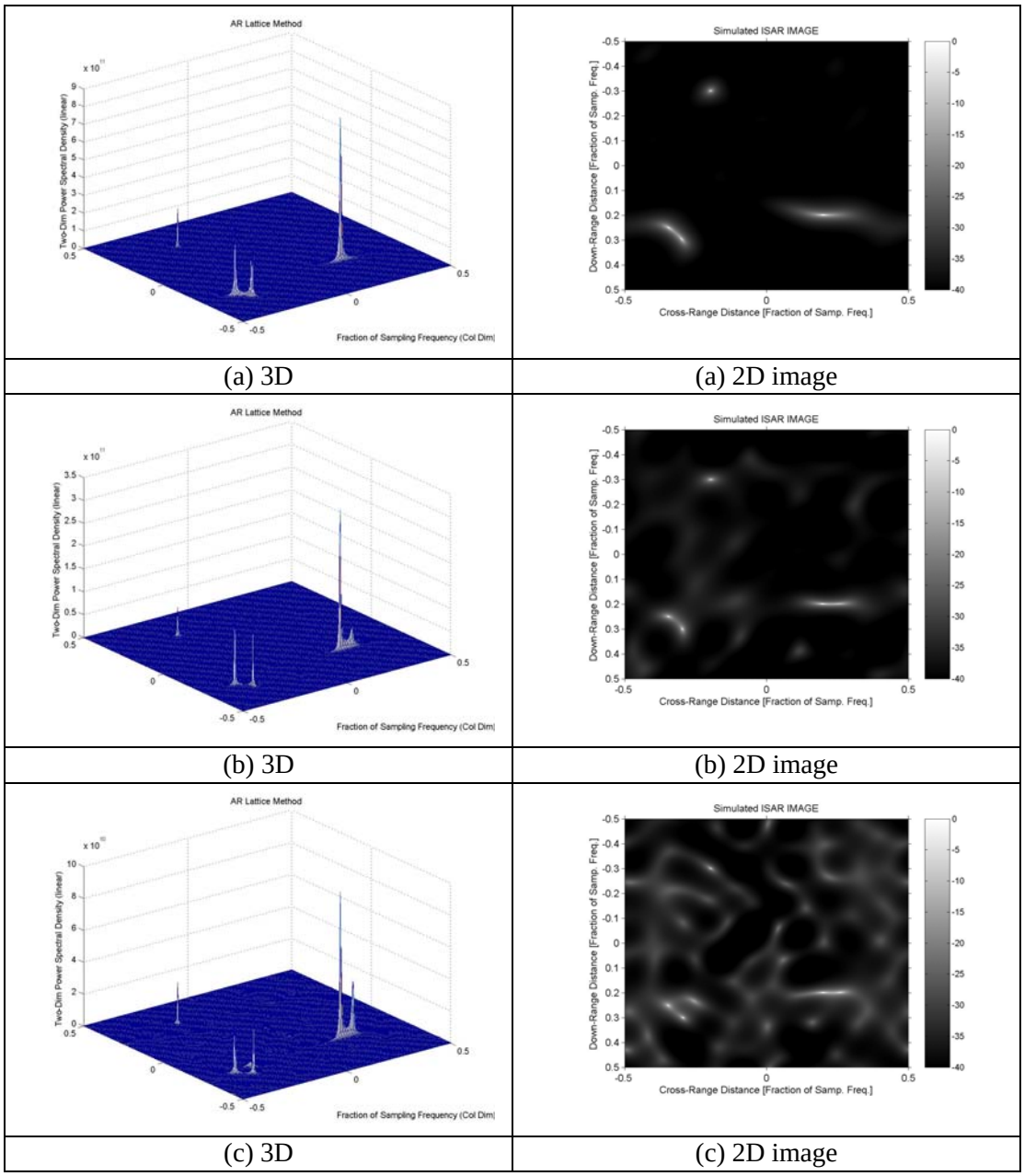
Sequentially minimizing the sum of the squared errors has the interesting property of guaranteeing that the reflection coefficients will be bounded by one in magnitude and therefore, the model will be stable.

There are several studies of two-dimensional linear prediction methods in the literature. It was shown in [25] that a solution to the 2-D linear prediction problem can be derived by relating it to the multichannel linear prediction problem and applying the multichannel Levinson algorithm. This connection led to several algorithms for solving for the 2-D linear prediction parameters [25, 26]. A fast computational algorithm that is essentially an extension of the original 1-D Burg algorithm, and that exploits the matrix structure inherent in the 2-D Linear Prediction (LP) problem, has been introduced in [27].

Several papers have also attempted to apply 2-D linear prediction to synthetic aperture radar imagery and computed medical imagery [28, 29].

The following analysis uses the fast computational 2-D lattice algorithm of [27] to compute enhanced simulated and turntable ISAR imagery. This algorithm estimates the first and fourth quadrant 2-D AR parameters from the 2-D data using a least squares method and the 2-D Levinson algorithm. Further simplifications are made by exploitation of the doubly Toeplitz block matrix structure that exists in the problem. The reader is encouraged to read chapter 16 of [22] for a more detailed treatment of 2-D AR modeling.

The results of using the 2-D lattice algorithm to compute imagery from the simulated ISAR data is shown in Figure 3.8. In chapter 4 the imagery produced by this method will be shown to have increased resolution enhancement over the classical estimators of chapter 2 and the AR Yule Walker method. The danger with the 2-D (1-D also) lattice algorithm is the possibility of introducing artifacts such as line splitting in the imagery at higher model orders. However, it has been observed that this can be a problem to some degree with all of the modern 2-D spectral estimators presented in this thesis. A rule of thumb is to never choose a model order that is larger than twice the available data record length, and ideally never exceed one third the available data record length. The problem of properly selecting the AR model order for 1-D AR spectral estimation is well documented [22], and approaches to model order selection for 2-D AR models have been presented in [30]. However, model order selection for ISAR imagery enhancement using the 2-D lattice algorithm has not been thoroughly explored, and at this point remains somewhat subjective.




(a) 3D | (a) 2D image

(b) 3D | (b) 2D image

(c) 3D | (c) 2D image

Figure 3.8 Simulated ISAR images of five point targets with additive white noise, 16x16 data sample set. AR Lattice method with model order (p,q) for both columns and rows (a) AR(3,3) (b) AR(5,5) (c) AR(7,7)

The results of applying the 2-D lattice algorithm to the T-72 Tank data set are shown in Figure 3.8. Even though the differences between the simulated images for the lattice method and that of previous methods are visually dramatic, the visual differences between the Yule-Walker method and the Lattice method imagery for the T72 tank is small. However, the qualitative analysis in chapter 4 will demonstrate that the lattice method does in fact produce imagery with certain characteristics consistent with the notion of higher resolution. It is important to keep in mind that the visual appearance of the ISAR imagery to the eye is not the best benchmark for determining imagery enhancement. Radar imagery can provide information via specific target signatures that can be exploited for target recognition and detection that are not necessarily exploited by the human perception system. These signatures can be unique such that computing systems can use them to identify the target. The task of quantifying the performance of each spectral estimation method for applications that require resolved signatures for target identification will be presented in chapter 4.

Visual enhancement can be seen in Figure 3.8 around the turret of the tank where scatterers are close together. In this region the lattice method does a better job resolving scattering centers than the Yule Walker method. This is one reason that the lattice method is more suitable than the previous methods for applications such as Automatic Target Recognition (ATR). However, the added resolution does come at the cost of a higher computational burden. The analysis in chapter 4 will attempt determine if the enhancement from the lattice method is worth the increase in computational cost from a comparison standpoint.

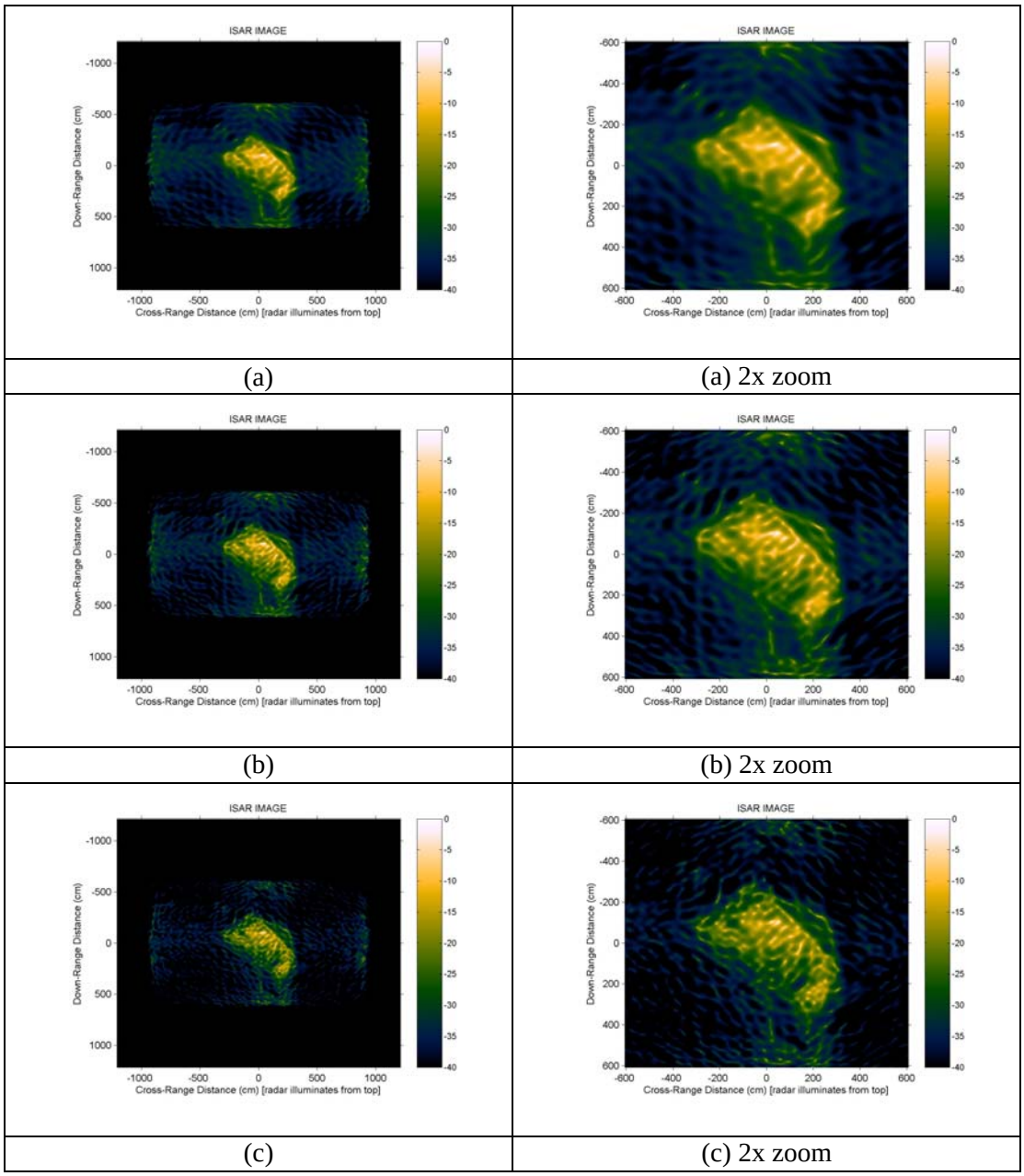



Figure 3.9 ISAR images of T-72 tank. AR Lattice method with model order (p,q) for both columns and rows (a) AR(30,30) (b) AR(40,40) (c) AR(50,50)

### 3.2.6 2-D COVARIANCE METHOD

The two-dimensional covariance method is also best understood by first exploring the one-dimensional covariance method. Like the lattice, or Burg algorithm, the covariance algorithm is based on a least squares approach and involves minimizing the sum of the squared errors from a forward or backward linear prediction error filter [31]. The difference for the covariance method is that no assumption is made about the unobserved data. There are several variations of this method, and the reader is pointed to [22] for a complete treatment.

A fast two-dimensional covariance algorithm has been developed in [32]. This algorithm exploits the near-to-doubly Toeplitz structure of the normal equations. As mentioned earlier the 2-D first quadrant, quarter plane linear prediction error filter is defined as

$$e^1\left[n_1, n_2\right] = \sum_{k_1=0}^{p_1} \sum_{k_2}^{p_2} a^1\left[k_1, k_2\right] x\left[n_1 - k_1, n_2 - k_2\right] \quad , \tag{3.12}$$

in which $a^1[0,0] = 1$ by definition. This equation can also be expressed in block vector form as

$$e^1\left[n_1, n_2\right] = \underline{a}^1 \underline{x}\left[n_1, n_2\right] \ , \tag{3.13}$$

where $\underline{x}^T[n_1, n_2] = \left(x[n_1, n_2] \ldots x[n_1 - p_1, n_2] \ldots x[n_1, n_2 - p_2] \ldots x[n_1 - p_1, n_2 - p_2]\right)$ (3.14)

is a data vector, and $\underline{a}^1 = \left(a^1[0] \quad a^1[1] \quad \ldots \quad a^1[p_2]\right)$ (3.15)

is a block vector with vector elements $a^1[p] = \left(a^1[0, p] \ a^1[1, p] \ \ldots \ a^1[p_1 p]\right)$ (3.16)

for $0 \le p \le p_2$.

As mentioned at the beginning of this section, the 2-D least squares normal equations for the 2-D covariance method are obtained by using the data available over the intervals $0 \le n_1 \le N_1 - 1$ and $0 \le n_2 \le N_2 - 1$. No assumption is made about data outside the observed

data length. Therefore, the summation for the squared errors only extends through the data observation interval and is expressed as

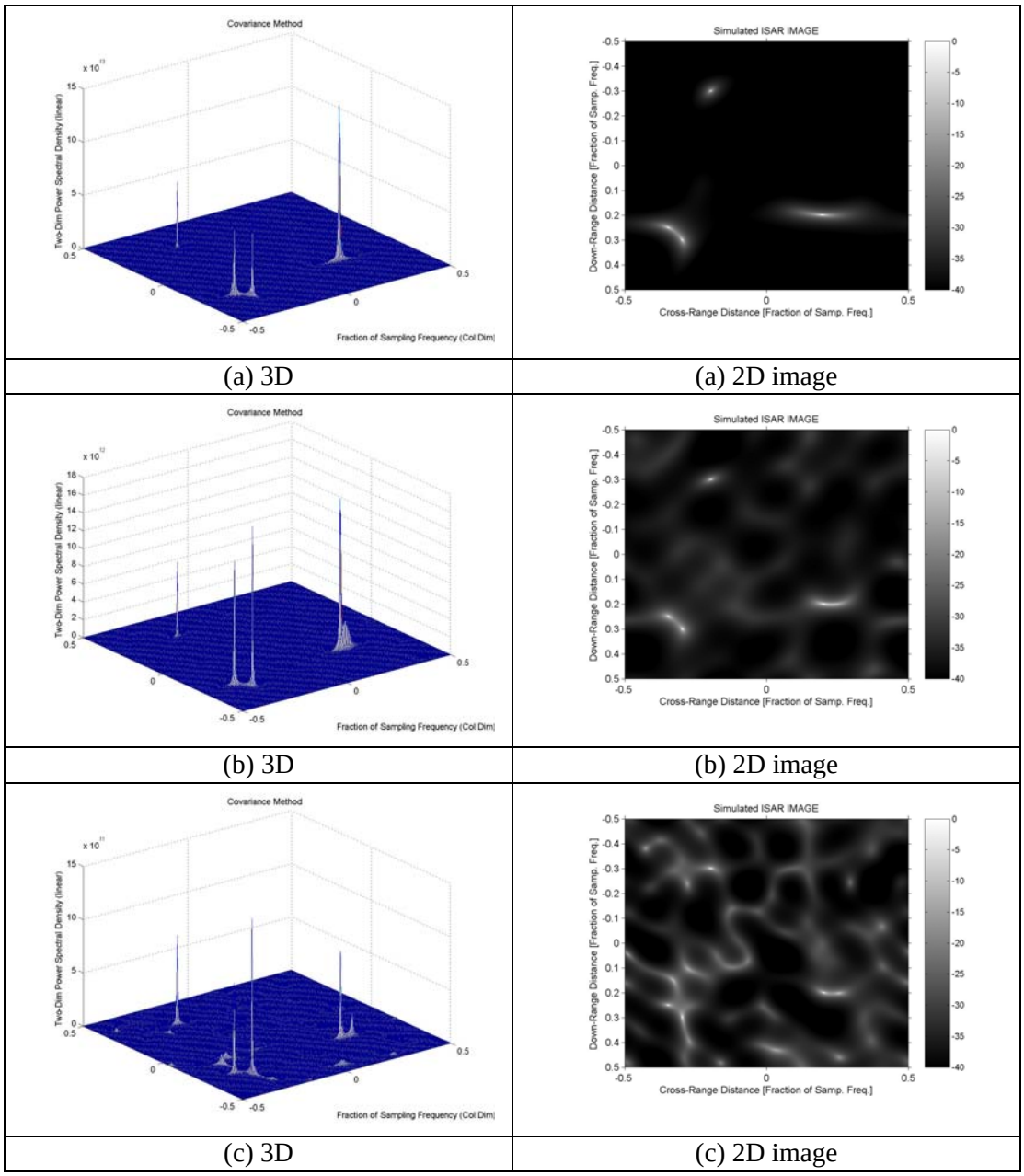
$$\begin{aligned}\rho^1 &= \sum_{n_1=p_1}^{N_1-1}\sum_{n_2=p_2}^{N_2-1}\left|e^1[n_1,n_2]\right|^2 \\ &= \underline{a}^1\,\underline{R}\,\underline{a}^{1H}\end{aligned} \quad . \tag{3.17}$$

The block RCS matrix in Equation 3.17 can be expressed as

$$\begin{aligned}\underline{R} &= \sum_{n_1=p_2}^{N_1-1}\sum_{n_2=p_2}^{N_2-1}\underline{x}[n_1,n_2]\underline{x}^H[n_1,n_2] \\ &= \underline{X}\,\underline{X}^H \\ &= \begin{pmatrix} R[0,0] & R[0,1] & \cdots & R[0,p_2] \\ R[1,0] & R[1,1] & \cdots & R[1,p_2] \\ \vdots & \vdots & \ddots & \vdots \\ R[p_2,0] & R[p_2,1] & \cdots & R[p_2,p_2] \end{pmatrix}\end{aligned} \quad , \tag{3.18}$$

where $R[i,j]$ are matrix elements and $\underline{X}$ is a block-Toeplitz 2-D data matrix made up of block elements that are 2-D Toeplitz data matrices. After the total squared error is minimized a set of least squares normal equations are found that can be solved using a special variant of the multichannel covariance algorithm [33]. The solution results in the filter parameters that are then used to compute the PSD from Equation 3.8.

The results of using the 2-D covariance algorithm to compute simulated ISAR imagery is shown in Figure 3.9. For low orders, four targets are easily recognized and increased sharpening of the imagery, compared to the non-parametric methods, is achieved. However, the fifth target that is weak and close to the sidelobes of another stronger target is not noticeable in the imagery until the model order is increased to be equal or greater than the number of targets. At these higher order values it becomes difficult to determine if the target is real or an artifact of the estimation model.




| | |
|---|---|
| (a) 3D | (a) 2D image |
| (b) 3D | (b) 2D image |
| (c) 3D | (c) 2D image |

Figure 3.10 Simulated ISAR images of five point targets with additive white noise, 16x16 data sample set. LP Covariance method with model order (p,q) for both columns and rows (a) LP(3,3) (b) LP(5,5) (c) LP(7,7)

The results of the modified covariance algorithm used to compute the T-72 tank imagery are shown in Figure 3.11. As with the other AR methods presented so far the covariance method results in imagery that does not suffer from side-lode artifacts and speckle.

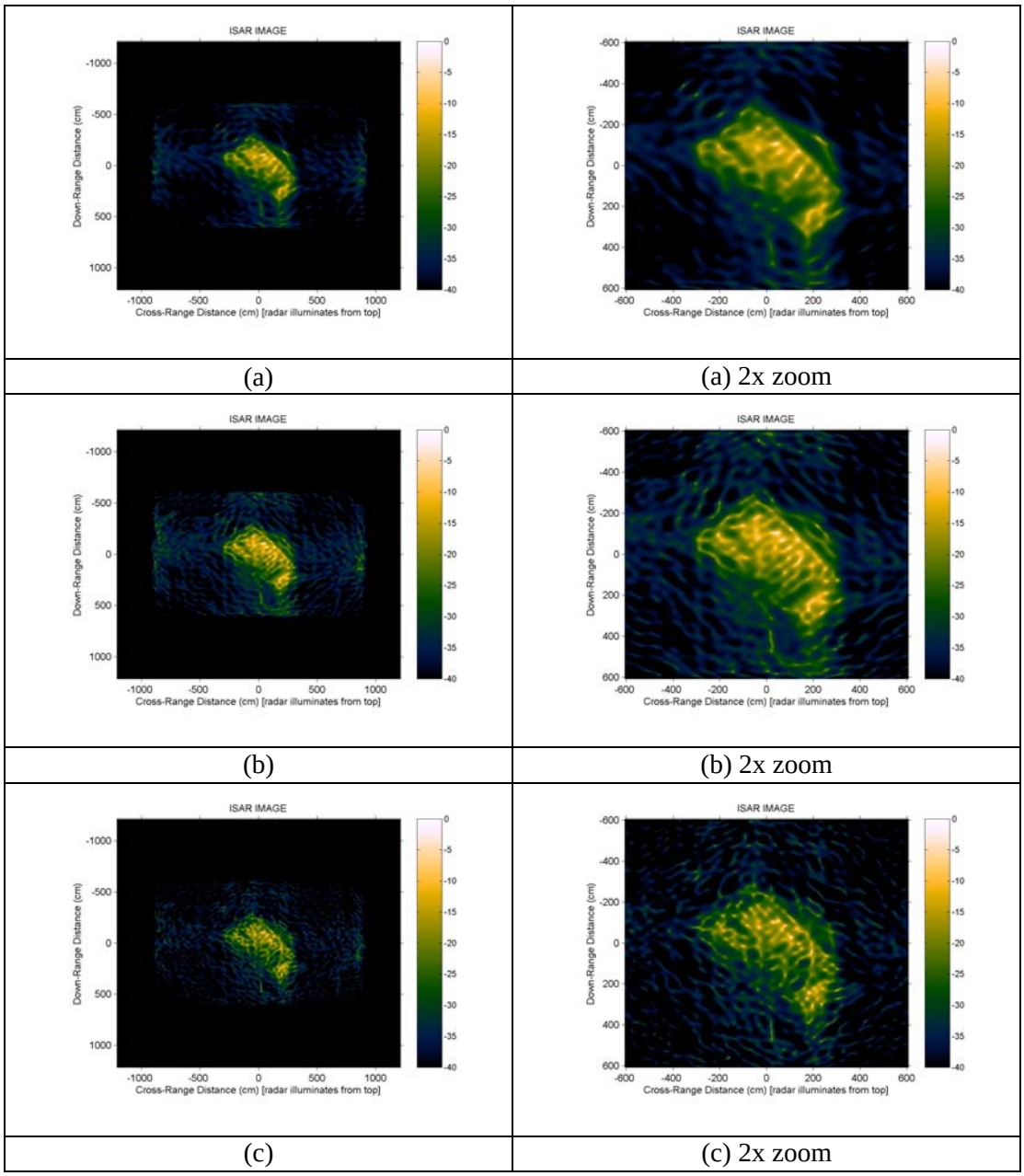



Figure 3.11 ISAR images of T-72 tank. LP Covariance method with model order (p,q) for both columns and rows (a) LP(30,30) (b) LP(40,40) (c) LP(50,50)

There is also an obvious increase in resolution over the AR Yule-Walker method and better localization of the strong point scatterers in the turret region. At a model order of 40 for both rows and columns, the tank turret is easily recognizable and the attached feature on the rear of

the tank shows more detail. The fundamentals of the covariance method have also been used to develop methods to reduce speckle in polarimetric SAR imagery [34].

### 3.2.6 2-D MODIFIED COVARIANCE METHOD

The two-dimensional modified covariance method used for the analysis in this thesis was introduced in [35] and experimental results of a modified covariance method applied to radar imagery can be found in [36]. The use of the modified covariance method to enhance radar imagery has received critical review [23] due to the fact that, unlike the Burg method, the prediction coefficients calculated using the modified covariance method are not guaranteed to be stable. It is useful to explore the 1-D modified covariance method in order to understand the 2-D method. A thorough treatment of the 1-D modified covariance method can be found in [22] and a fast computational algorithm was introduced in [35]. A simple introduction is presented here for completeness.

The modified covariance uses both the forward and backward linear prediction statistics to generate additional error points. This is possible because the forward linear prediction coefficients and the backward linear prediction coefficients are complex conjugates of one another. Like the covariance method, the squared errors over the available data can be computed by

$$\rho_p^{fb} = \frac{1}{2}\sum_{n=p+1}^{N}\left|e_p^f[n]\right|^2 + \frac{1}{2}\sum_{n=p+1}^{N}\left|e_p^b[n]\right|^2 \quad . \tag{3.19}$$

Minimizing the expression for the squared errors leads to a set of normal equations that can be solved directly using the Cholesky algorithm [22] or indirectly using the fast computational algorithm developed in [37]. An extension of this 1-D algorithm for 2-D data sets has been developed in [35]. Like the 2-D covariance method, it is developed from the assumption that

the 2-D data is available only over the intervals $0 \le n_1 \le N_1 - 1$ and $0 \le n_2 \le N_2 - 1$. The 2-D linear prediction errors can be computed by

$$e^i\left[n_1, n_2\right] = \sum_{k_1=0}^{p_1} \sum_{k_2}^{p_2} a^i\left[k_1, k_2\right] x\left[n_1 - k_1, n_2 - k_2\right] \quad , \tag{3.20}$$

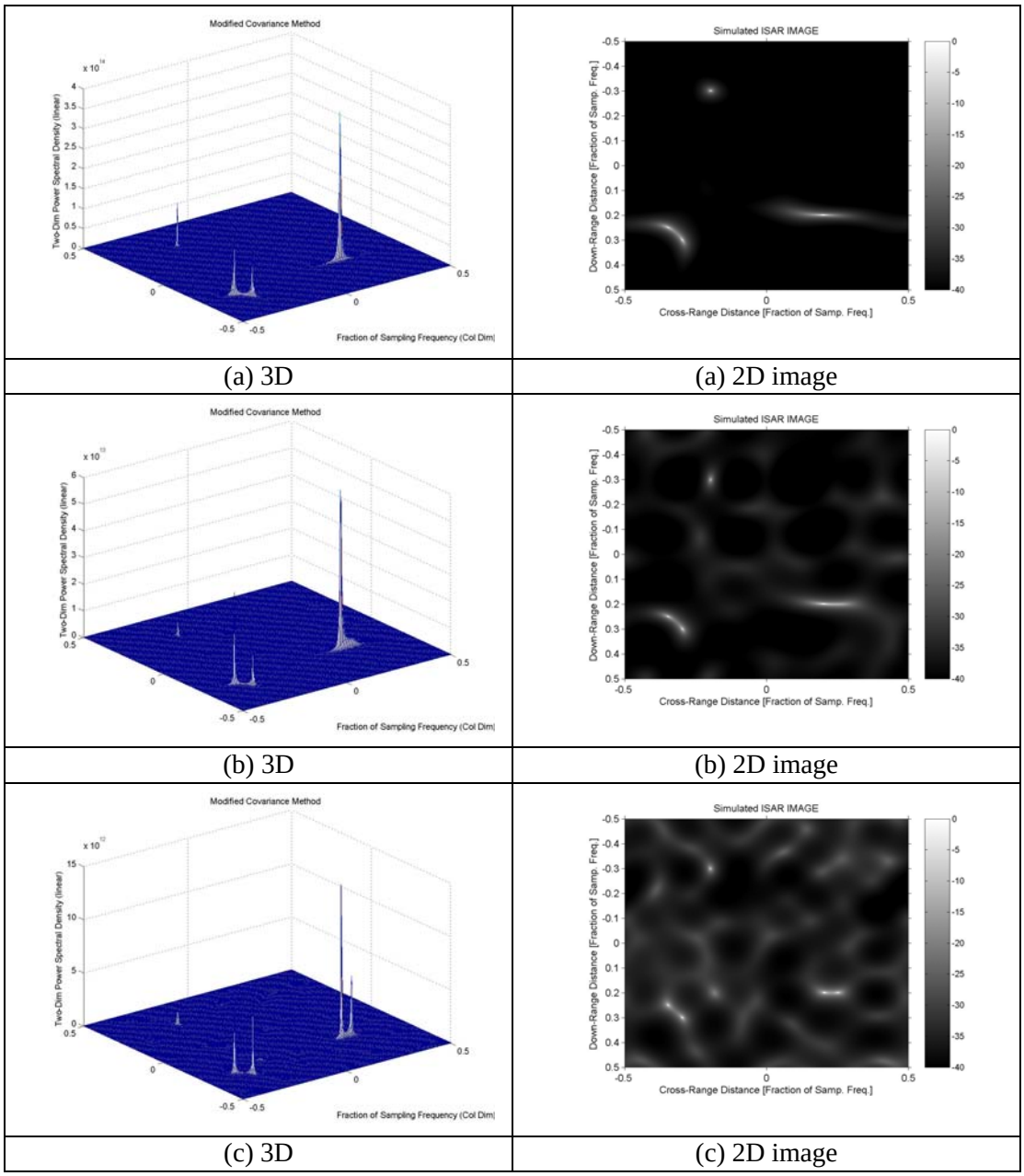
in which $a^1[0,0] = 1$ by definition. For the modified covariance method the sum of squared errors becomes

$$\begin{aligned} \rho^{1+3} &= \sum_{n_1=p_1}^{N_1-1} \sum_{n_2=p_2}^{N_2-1} \left|e^1[n_1, n_2]\right|^2 + \left|e^3[n_1, n_2]\right|^2 \\ &= \underline{a}^1 \underline{R} \underline{a}^{1H} \end{aligned} \quad , \tag{3.21}$$

assuming $\underline{a}^{3*} = \underline{a}^1$, which is valid if the 2-D autocorrelation sequence is known. The details of the fast computational algorithm are beyond the scope of this thesis and can be found in greater detail in [35]. Once the filter parameters are computed using this fast algorithm, the parameters are used to compute the 2-D PSD using Equation 3.8.

Results of using the 2-D modified covariance method to compute imagery from the simulated ISAR data are shown in Figure 3.12. It is clear that the modified covariance method does very well at sharpening the response for all five targets when the model order is slightly larger than the number of targets. The method seems to suffer less from artifacts such as line splitting and strong localized clutter areas than the previous methods discussed.

The results for the T72 tank data are shown in Figure 3.13. The results are somewhat disappointing considering the good results with the simulated data. There does not seem to be any improvement over the 2-D covariance method and the clutter levels are actually higher. The 2-D modified covariance method requires more computation time as compared to the 2-D covariance method, lattice method, or Yule Walker method.




(a) 3D | (a) 2D image

(b) 3D | (b) 2D image

(c) 3D | (c) 2D image

Figure 3.12 Simulated ISAR images of five point targets with additive white noise, 16x16 data sample set. LP Modified Covariance method with model order (p,q) for both columns and rows (a) LP(3,3) (b) LP(5,5) (c) LP(6,6)

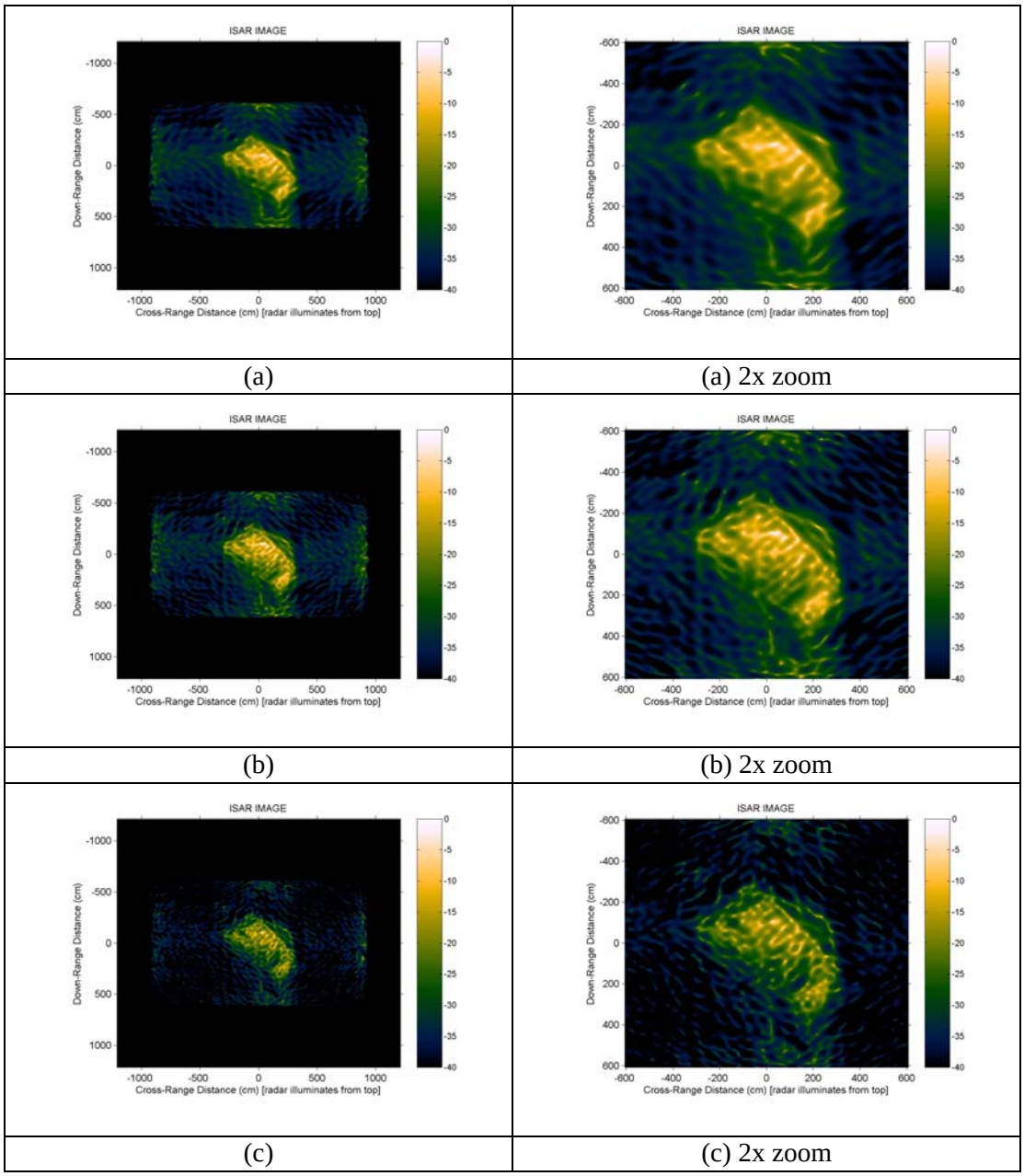



Figure 3.13 ISAR images of T-72 tank. LP Modified Covariance method with model order (p,q) for both columns and rows (a) LP(30,30) (b) LP(40,40) (c) LP(50,50)

### 3.2.7 2-D MINIMUM VARIANCE METHOD

The minimum variance (MV) method creates a spectral estimate by minimizing the variance of the output of a bandpass filter that adapts to the processed input data spectral

content. The MV is a spectral estimator in that it produces a representation of the relative spectral component strengths of the input data over frequency. It differs from the classical spectral estimators in that it does not produce a power spectral density (PSD) function that has an area under the function representing the total power in the process [22]. However, it does produce peaks that are linearly proportional to the power of the point targets present in the spectrum. The minimum variance method of spectral estimation has been recognized as a useful tool for enhancing radar imagery [38], and there are several variations of the basic concept in the literature [39,40].

A thorough treatment of the 1-D minimum variance method of spectral estimation can be found in [22], where it is shown that the MV spectral estimator is given by

$$P_{MV}(f) = \frac{T}{e^H(f) R_p^{-1} e(f)} \ , \tag{3.22}$$

where

$$e(f) = \begin{pmatrix} 1 \\ \exp(j2\pi fT) \\ \vdots \\ \exp(j2\pi fpT) \end{pmatrix} , \tag{3.23}$$

and $R_p^{-1}$ is the estimated or known autocorrelation matrix inverse. There is also a relationship between the MV spectral estimator and the AR spectral estimator. This relationship has been developed for both the frequency domain and the time domain [41]. The frequency domain connection can be expressed as,

$$\frac{1}{P_{MV}(f)} = \sum_{k=0}^{p} \frac{1}{P_{AR}(k, f)} \ , \tag{3.24}$$

which relates the reciprocal of the MV estimator to the average of the reciprocals of the AR estimators from order $k = 0$ to $k = p$. This relationship to the average of the AR estimators explains why an MV estimate can result in lower resolution imagery than AR estimation for a given input process. However, the MV estimate will have a greater resolution than classical spectral estimators, and will produce a spectral estimate with a lower variance than an AR estimator will. It is this combination of characteristics that makes it a popular choice for radar imagery enhancement.

The 2-D minimum variance algorithm used for the work presented in this section was developed in [42]. It is based on the fact that the 2-D data covariance matrix has special structure that can be exploited to arrive at a closed form for the inverse covariance matrix. The 2-D minimum variance spectral estimator for first quadrant support can be expressed as

$$P(f_1, f_2) = \frac{1}{\bar{e}(f_1, f_2)\bar{R}^{-1}\bar{e}^{H}(f_1, f_2)} \ , \tag{3.25}$$

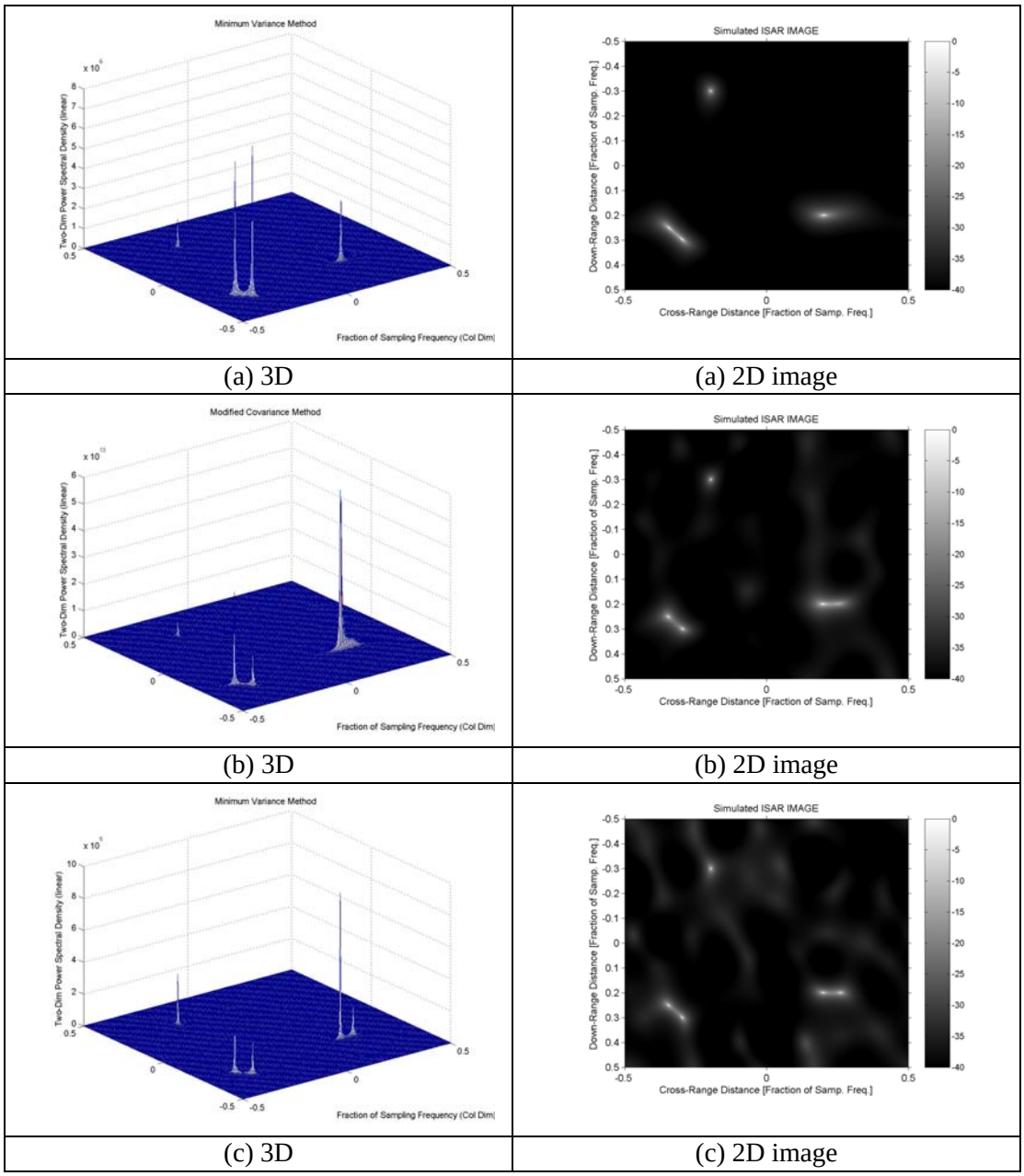
where $\bar{R}^{-1}$ is the inverse of the doubly Toeplitz autocorrelation matrix and

$$\bar{e}(f_1, f_2) = [e(f_1), \exp(j2\pi f_2)e(f_1), \cdots, \exp(j2\pi f_2[p_2])e(f_1)] \tag{3.26}$$

and

$$e(f_1) = [1, \exp(j2\pi f_1), \cdots, \exp(j2\pi[p_1]f_1] \quad . \tag{3.27}$$

The results of applying the 2-D MV algorithm to the simulated ISAR data are shown in Figure 3.14. At model order 6 for rows and columns, all five targets are resolved and the clutter is smoothed or averaged as compared to the modified covariance and covariance methods. This makes it easier to detect the targets and less likely to confuse estimates of the clutter with that of the actual point targets.




| | |
|---|---|
| (a) 3D | (a) 2D image |
| (b) 3D | (b) 2D image |
| (c) 3D | (c) 2D image |

Figure 3.14 Simulated ISAR images of five point targets with additive white noise, 16x16 data sample set. Minimum Variance with model order (p,q) for both columns and rows (a) MV(3,3) (b) MV(5,5) (c) MV(6,6)

The results for the T-72 tank data are shown in Figure 3.15. Even at lower model orders, the MV method yields good quality imagery that resolves many features of the tank that were not visually resolvable with the classical spectral estimators. The imagery from the

2-D MV method for higher model orders begins to take on the appearance of electro-optic (EO) imagery. The MV method does not have the theoretical resolution of the lattice method or the covariance based methods, but it performs well with measured data and real collections [38].

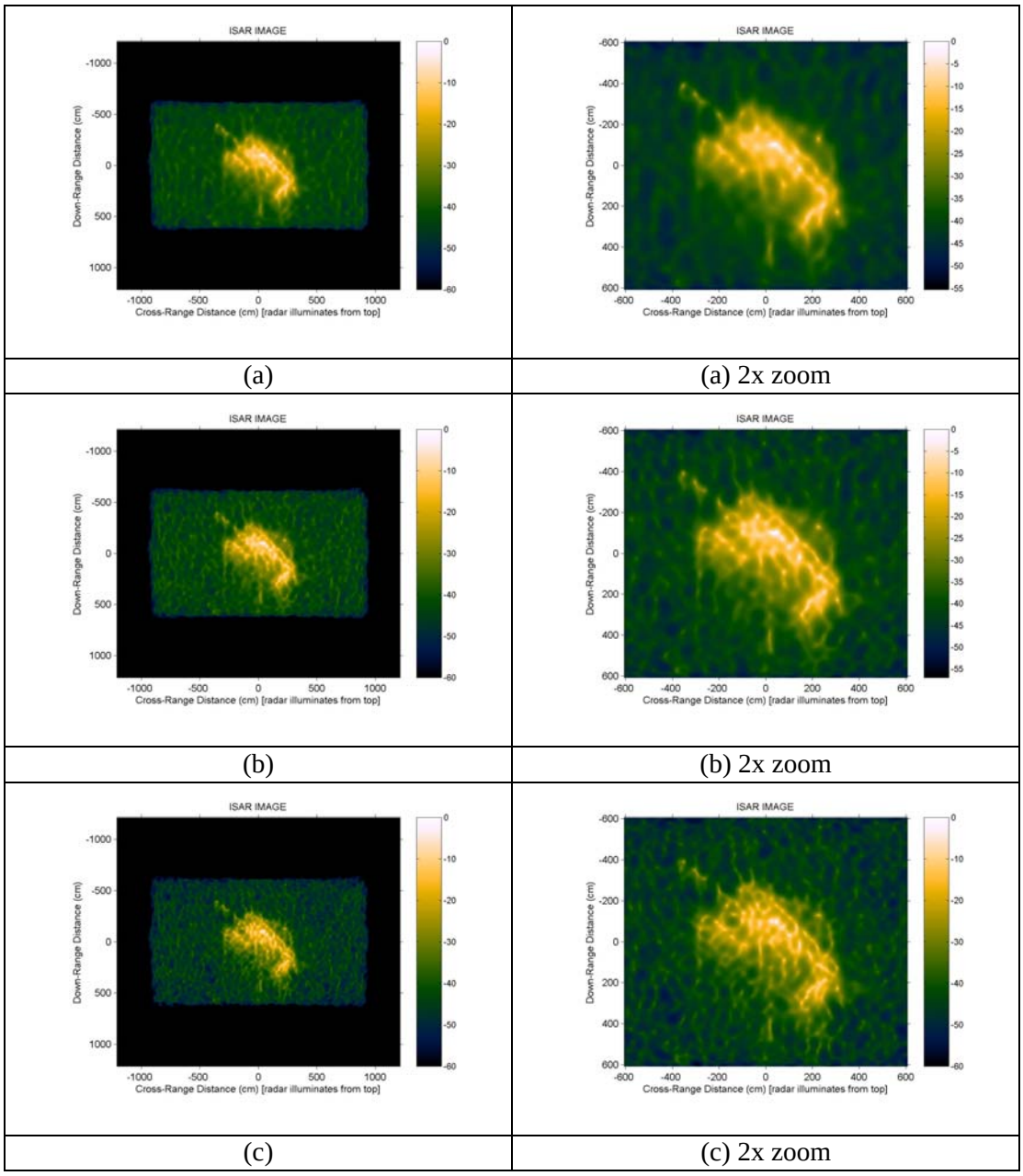



Figure 3.15 ISAR images of T-72 tank. Minimum Variance method with model order (p,q) for both columns and rows (a) LP(30,30) (b) LP(40,40) (c) LP(50,50)

### 3.2.7 HYBRID SPECTRAL ESTIMATION METHODS

All of the modern 2-D spectral estimators presented in this thesis can be used to enhance radar imagery using only their 1-D counterpart. This technique is called hybrid spectral estimation, and the diagram below outlines the general steps of the technique. First, a 1-D FFT is computed along all rows or columns. Then a modern 1-D spectral estimator is computed along the orthogonal row or column direction. This technique is useful if increased resolution is only needed in the range or cross-range dimension.

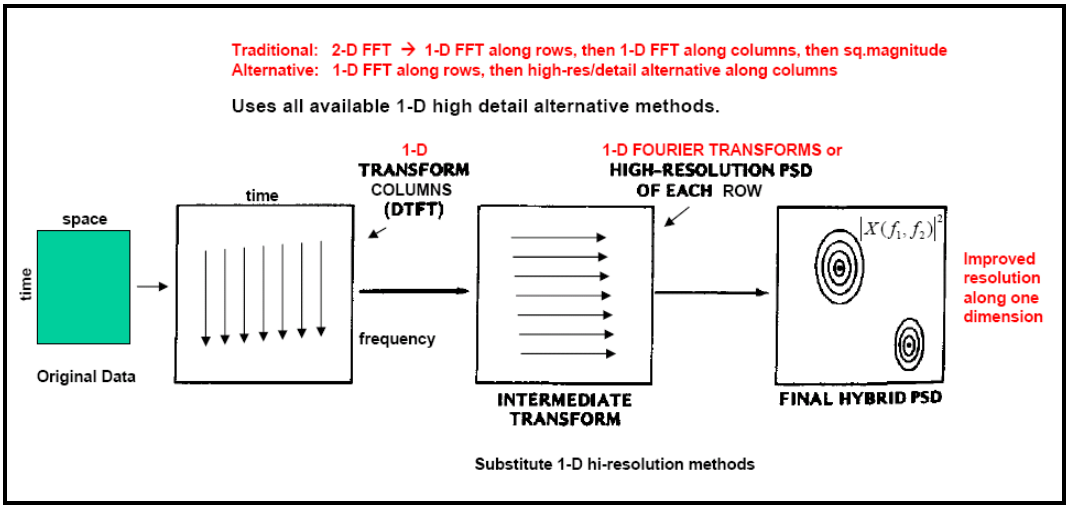



Figure 3.16 Hybrid spectral estimation method [Marple ECE679 Class Notes]

Hybrid methods can be extremely useful when computation time is limited or computational resources are not adequate to compute 2-D methods.

## 3.3 OTHER HIGH RESOLUTION METHODS

There are many other high-resolution methods for enhancing radar imagery that have been developed and presented in the literature. A good review of many of the modern techniques can be found in [38]. Some more recent methods include the relaxed-based

parametric methods and the semi-parametric or SPAR methods [45]. Time-frequency analysis (TFA) can also be applied to ISAR imaging problems [43, 44]. It has advantages when trying to image objects that suffer from multi-path effects or that need motion compensation in order to produce well-focused imagery. Figure 3.17 shows ISAR images created using time-frequency methods. The images are formed from real ISAR data of a B-727 collected from a stepped-frequency radar operating at 9GHz, with a bandwidth of 150 MHz. For each pulse, 128 complex range samples were saved. Motion compensation and range processing were applied to the data. The left image in Figure 3.17 is a radar image formed by using the short-time Fourier transform (STFT) with a time-frequency method. The image on the right was computed using a high-resolution linear prediction method. High-resolution spectral estimation techniques are useful with time-frequency methods in order to achieve enhanced imagery [43]. The right image of Figure 3.17 is an example of enhanced resolution in one-dimension (azimuth). There are many other uses of spectral estimation techniques applied with TFA to extract features from radar signals and imagery that are being explored

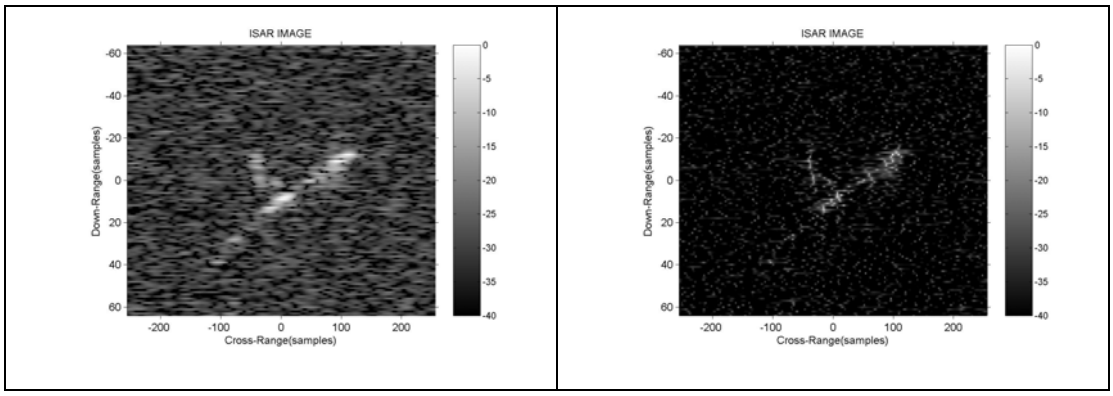



Figure 3.17 Time-frequency ISAR images of real B-727. Image computed by STFT technique (Left). Image computed by high-resolution linear prediction technique (Right).

# CHAPTER FOUR

## 4.1 RESOLUTION ENHANCEMENT

It is useful to quantify the resolution enhancement gained by applying a particular spectral estimation algorithm to the image formation step. In this section an assessment of the resolution increase and variance reduction achieved by applying modern spectral estimators to the T-72 tank data will be made. The motivation for this assessment is to determine if modern spectral estimators can enhance the ISAR imagery for applications such as ATD/ATR, where strong and stable signatures are necessary. An assessment of the dynamic range increase achieved by the top performing methods will also be made.

Figure 4.1 shows results from all of the image formation methods (not including time-frequency methods) presented in this thesis. In order to evaluate the resolution and variance characteristics from each image formation method, a 1-D cross section is taken through the imagery. The cross section is chosen to pass through two, closely spaced, point-like scatterers on the tank turret. Figure 4.2 shows the tank imagery with the 1-D cross sections marked. Figure 4.3 shows plots from the cross sections and provides a quantitative comparison of the performance of the 2-D classical spectral estimators (DFT, Welch periodogram, and Blackman-Tukey). It is easy to recognize that the 2-D DFT resolves the point scatterers better than the other two methods. However, this plot represents the 2-D DFT without any windowing applied to reduce sidelobes. Any application of a windowing function in the computation of the imagery will decrease the resolution. A comparison of several methods against the windowed 2-D DFT will be made at the end of this section. It is also recognized in Figure 4.3 that the 2-D Blackman-Tukey method has high sidelobes that can obscure the weaker scatterers on the target. The 2-D periodogram does reduce the variance, but at the cost of lower resolution.

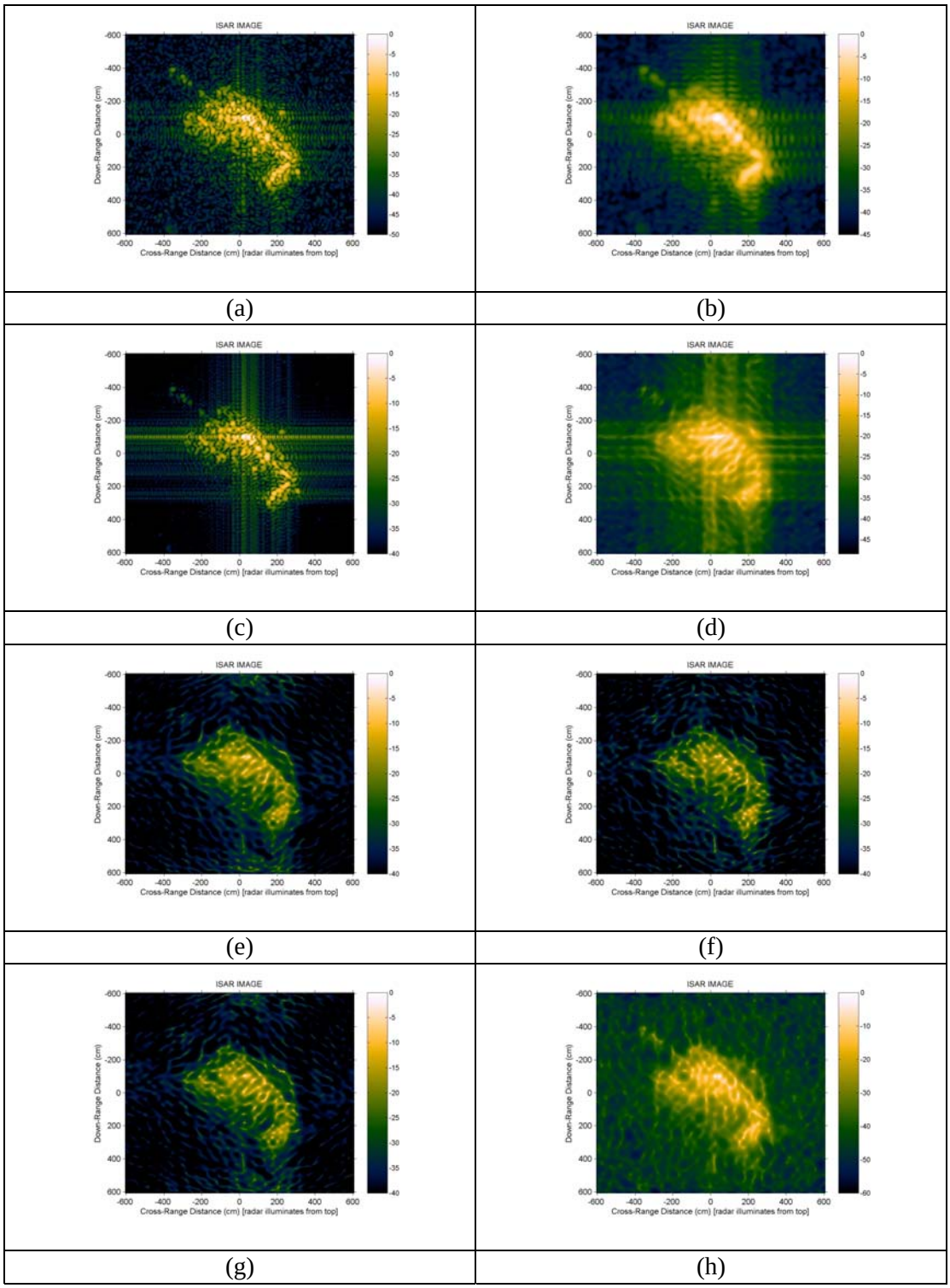
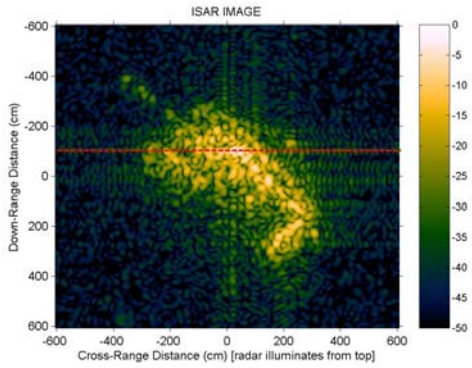
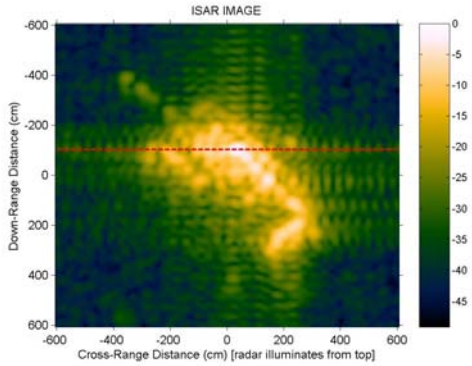


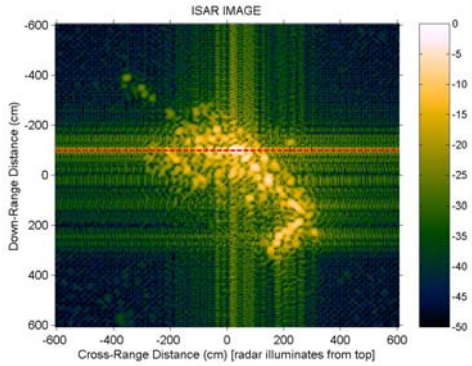

Figure 4.1 Imagery computed using all methods assessed in this thesis. (a) 2D DFT (b) 2D Welch periodogram (c) 2D Blackman-Tukey (d) 2D Yule-Walker (e) 2D lattice (f) 2D covariance (g) 2D modified covariance (h) 2D minimum variance




(a)




(b)




(c)

Figure 4.2 Images showing cross-sections through closely spaced point-like scatterers on T-72 turret. (a) 2D DFT (b) 2D Welch periodogram (c) 2D Blackman-Tukey

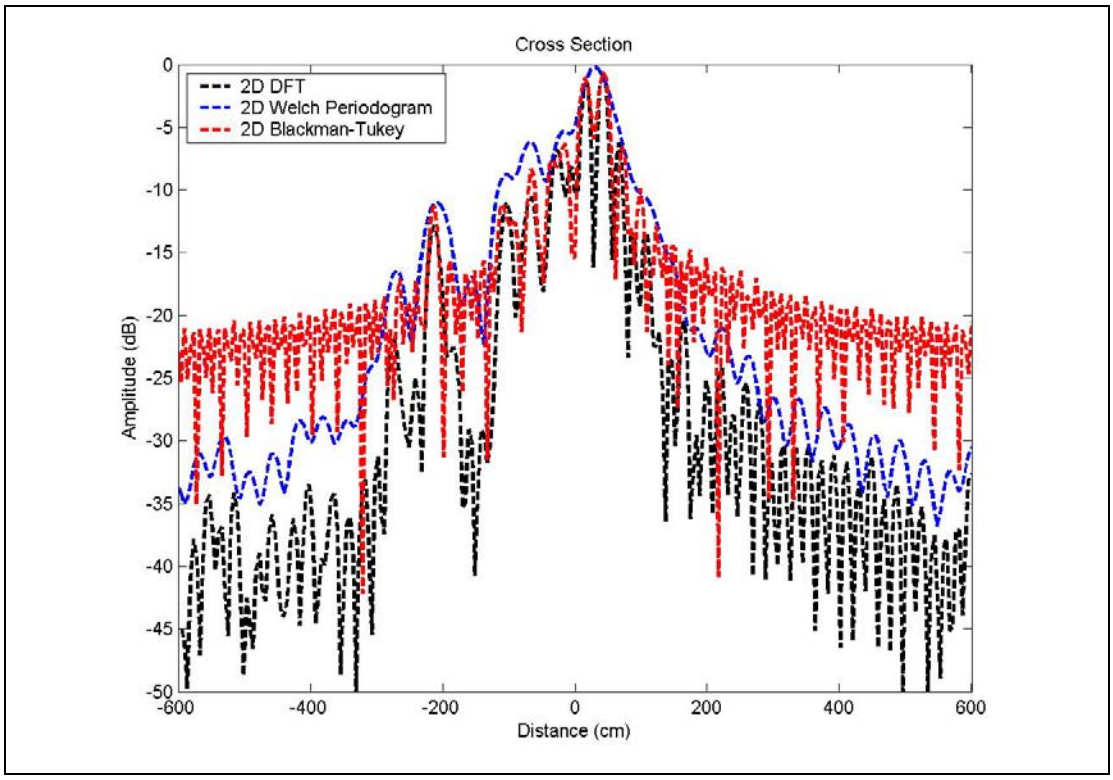



Figure 4.3 Cross sections through two point-like scatterers in T-72 ISAR imagery comparing resolution and variance performance of classical 2D spectral estimators.

Based off of the 2D simulated ISAR imagery results in chapter 3, we expect the modern spectral estimation methods to achieve higher resolution than the classical spectral estimation methods. Figure 4.4 shows the images computed using the modern spectral estimation methods along with the 1-D cross sections through the two closely spaced point scatterers on the tank turret market in red. Visually it can be determined that the Yule Walker method does not do a good job resolving the point scatterers on the T-72 turret. However, it is difficult to visually determine which of the other modern spectral estimation methods resolves the point scatterers the best. It is also not possible to determine visually if there is any dynamic range improvements by using one of the modern spectral estimators.

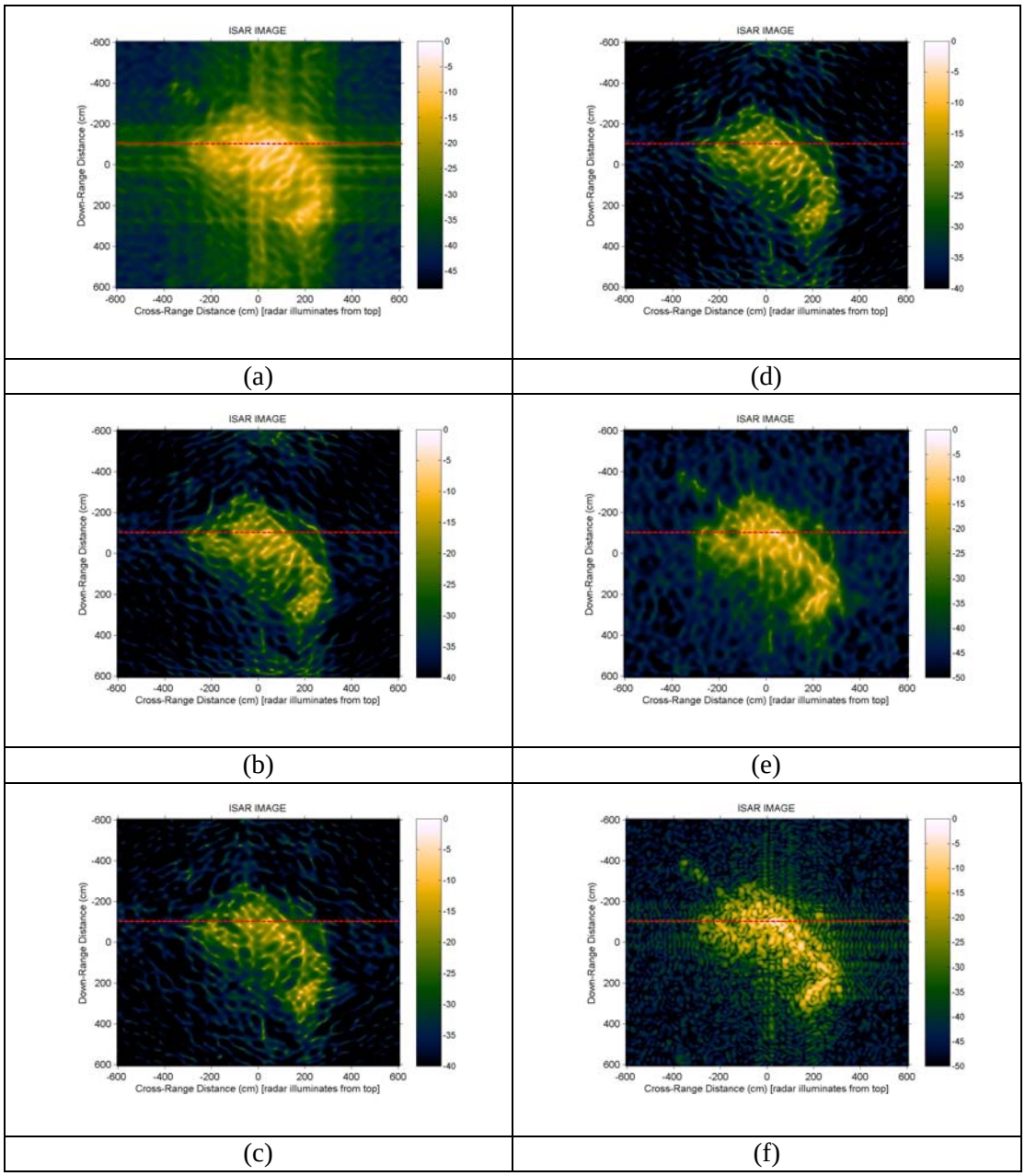



Figure 4.5 Cross sections through two point-like scatterers in T72 ISAR imagery comparing resolution and variance performance of modern 2D spectral estimators. (a) 2D Yule-Walker (b) 2D lattice (c) 2D covariance (d) 2D modified covariance (e) 2D minimum variance (f) 2D DFT

Figure 4.6 shows the 1-D plots from the cross sections in the imagery of Figure 4.5. The plots provide a quantitative comparison of performance for the 2-D modern spectral

estimators. It is easy to see that the 2-D Yule-Walker method has a low variance but very poor resolution. The two point scatterers are not resolved using this method, and the dynamic range is only about 30dB. The 2-D covariance and 2-D modified covariance methods also do not resolve the two closely spaced scatterers. However, they both provide a lower variance estimate than the 2-D DFT, and they sharpened responses of isolated strong scatterers. The 2-D minimum variance method achieves the best performance, providing both a low variance and high resolution. There is also an improvement in dynamic range over the other classical and modern spectral estimation methods. Figure 4.7 and 4.8 show the cross section plots from the three spectral estimation methods that perform the best with the T-72 tank data. These methods include the 2-D DFT, 2-D lattice and 2-D minimum variance. Figure 4.8 shows the 2-D DFT with sidelobe suppression weighting applied to be consistent with real applications.

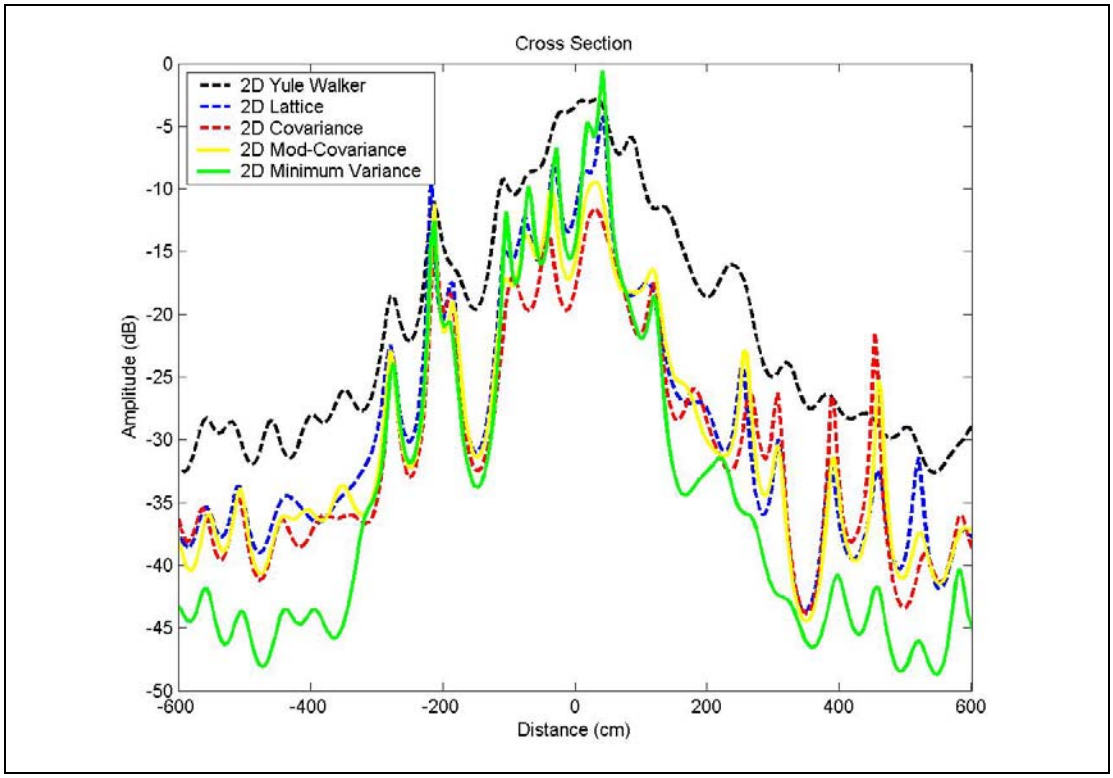



Figure 4.6 Cross sections through two point-like scatterers in T-72 ISAR imagery comparing resolution and variance performance of classical 2D spectral estimators.

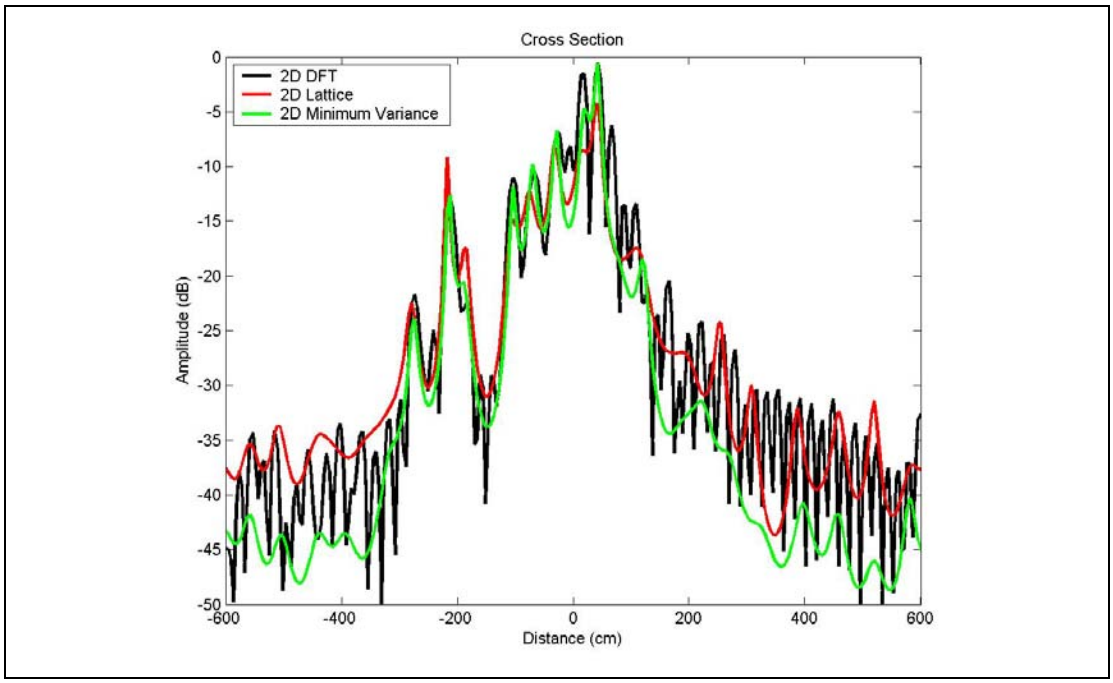



Figure 4.7 Cross sections through two point-like scatterers in T-72 ISAR imagery comparing resolution and variance performance of three best performing 2D spectral estimators: DFT, lattice, and minimum variance

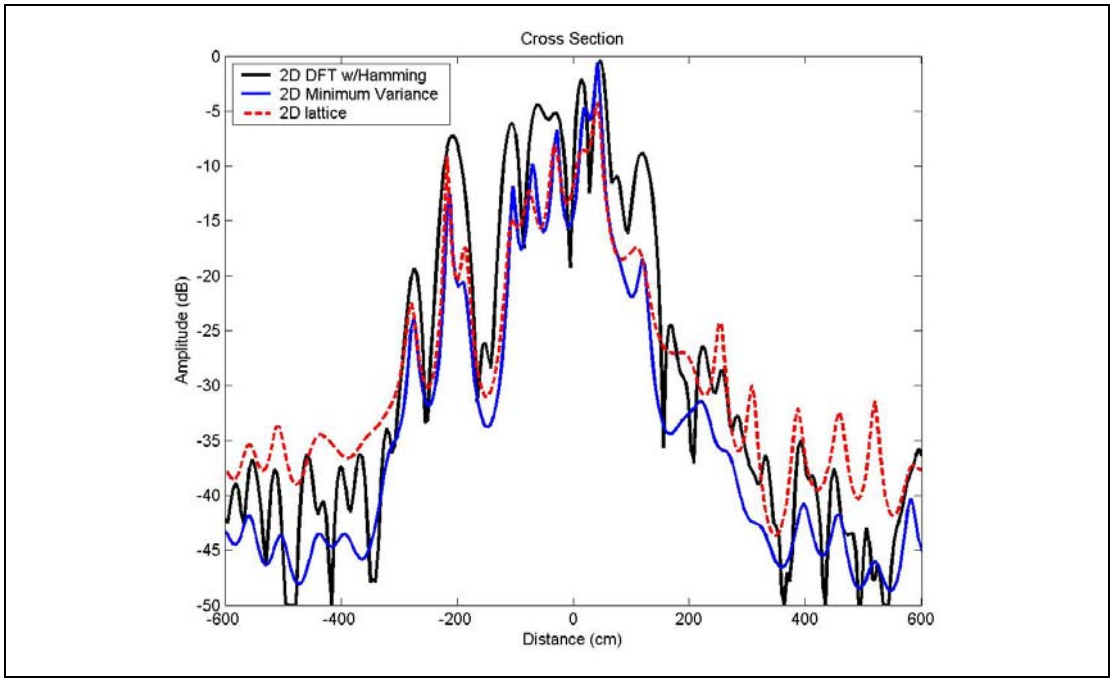



Figure 4.8 Cross sections through two point-like scatterers in T-72 ISAR imagery comparing resolution and variance performance of three best performing 2D spectral estimators: DFT with Hamming window, lattice, and minimum variance

It is easy to see that the minimum variance method achieves a sharpened response and lower variance than the 2-D DFT. It also achieves an increase in dynamic range. Both modern spectral estimation methods resolve target structure that is below the sidelobe levels when the 2-D DFT is employed. The increase in dynamic range and reduced variance makes the 2-D minimum variance especially useful for automatic target recognition (ATR) [38].

## 4.2 CONCLUSION

Several classical and modern spectral estimation methods were introduced and used to compute enhanced ISAR imagery from both simulated and real ISAR data sets. The topic of ISAR image formation was explored, starting from the geometric matched filter and then deriving the 2-D DFT method. These methods were used as baselines to compare image formation methods based on classical and modern 2-D spectral estimation techniques. Several important principles of ISAR image formation were introduced, such as zero-Doppler clutter subtraction and polar-to-cartesian interpolation. In addition to the 2-D DFT, two classical 2-D spectral estimators were introduced, including the 2-D Welch periodogram and the 2-D Blackman-Tukey method. Several 2-D modern spectral estimators were also introduced, including the 2-D Yule-Walker method, the 2-D lattice method, the 2-D covariance and modified covariance methods, and the 2-D minimum variance method. Imagery from simulated and real data was computed using all of these spectral estimation methods and visually compared. Imagery from each image formation technique, using several different parameters, was presented. In addition to the qualitative visual assessment of the computed imagery, a quantitative analysis was made of the resolution, variance and dynamic range performance of each spectral estimation method. Using the T-72 data, an image was computed and a 1-D cross section through the image was plotted for each method. The cross

section was chosen to pass through two closely spaced scatterers on the tank turret. Plots of the cross sections allowed for a comparison between the methods of resolution, variance and dynamic range. The 2-D DFT outperformed both of the other classical spectral estimation methods by achieving good resolution and dynamic range. The 2-D Blackman-Tukey method achieved good resolution but poor dynamic range, and suffered from strong sidelobe interference. The 2-D periodogram method achieved lower resolution than the 2-D DFT at the cost of poor resolution. It was also discovered that the 2-D Yule Walker method achieved low variance at the cost of poor resolution. In fact, it achieved poorer resolution than any of the classical spectral estimation methods when applied to the real T-72 data. Among the modern 2-D spectral estimation methods, it was discovered that the lattice and minimum variance methods performed the best. Both methods achieved lower resolution and increased resolution over the 2-D DFT. The minimum variance method also achieved higher dynamic range and can be a useful technique for automatic target recognition (ATR).

A Matlab toolbox for computing ISAR imagery using the methods presented in this thesis was developed. Figure 4.8 outlines the functions available in the toolbox.

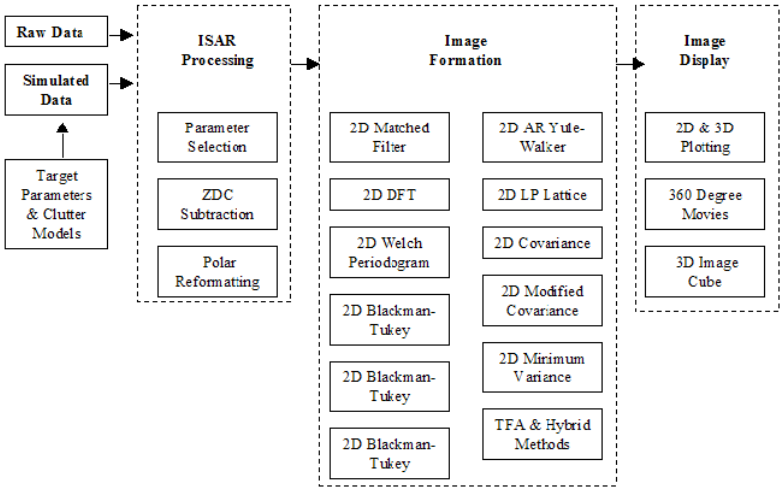



Figure 4.9 Block diagram showing the algorithms available in the Matlab toolbox developed for this thesis.

The toolbox can be used to compute imagery from raw data collected from a turntable range, such as the one at GTRI, or from an anechoic chamber. The toolbox can be easily expanded to include newly discovered algorithms and promising algorithms not covered in this thesis.

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# APPENDICES

## Appendix A. Inscribed rectangular grid within polar-annulus

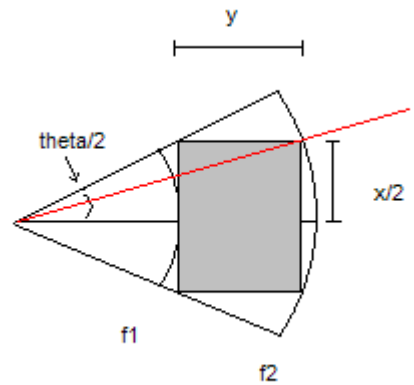


Figure A.1 Inscribed rectangular grid within azimuth aperture

from the diagram in Figure A.1 it is possible to get the following relationships

$$x = 2 f_1 \tan\left(\frac{\theta}{2}\right) \tag{A.1}$$

$$y = \sqrt{f_2^{\,2} - f_1^{\,2} \tan\left(\frac{\theta}{2}\right)} - f_1 \tag{A.2}$$

where $f_1$ and $f_2$ are the frequency extents of the data support region and $\theta$ is the azimuth extent. Since we want equal resolution in both dimensions after we have taken the Fourier transform of the phase history, we want x = y. Therefore, after some algebra we arrive at

$$5 f_1^{\,2} \tan^2\left(\frac{\theta}{2}\right) + 4 f_1^{\,2} \tan\left(\frac{\theta}{2}\right) - (f_2^{\,2} - f_1^{\,2}) = 0 \tag{A.3}$$

which is a quadratic equation. After solving Equation (A.3) for $\tan\left(\frac{\theta}{2}\right)$ we get

$$\tan\left(\frac{\theta}{2}\right) = \frac{-2 f_1^{\,2} + \sqrt{-f_1^{\,4} + 5 f_1^{\,2} f_2^{\,2}}}{5 f_1^{\,2}} \tag{A.4}$$

we can use this expression to solve for the dimensions of the inscribed rectangular grid.

$$x = \frac{\sqrt{-f_1^{\,2} + 5\,f_2^{\,2}} - f_1}{5} \tag{A.5}$$

and

$$\theta = 2\tan^{-1}\left(\frac{x}{f_1}\right) \tag{A.6}$$